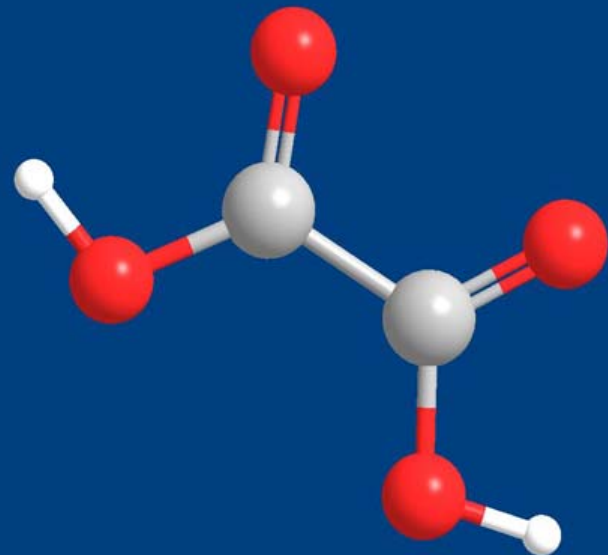
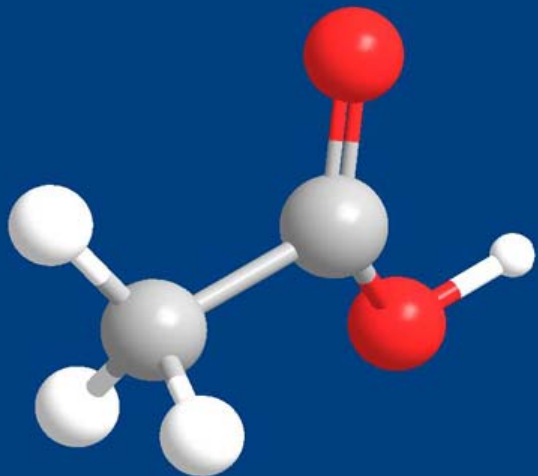




第十二章 羧酸及其衍生物

Carboxylic Acids and Carboxylic Acid Derivatives

厦门大学有机化学B

CONTENT

1

羧酸的结构、命名和物理性质

2

羧酸的化学性质

3

羧酸的制备

4

羧酸衍生物的命名和物理性质

5

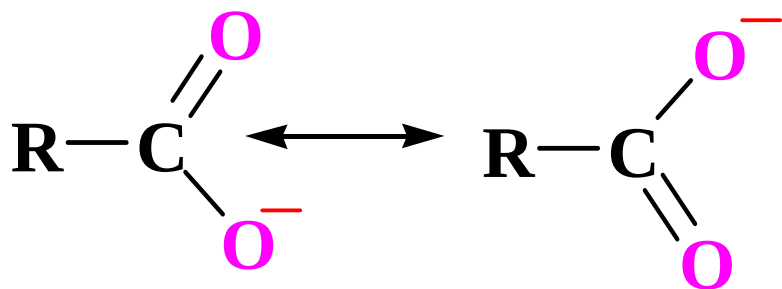
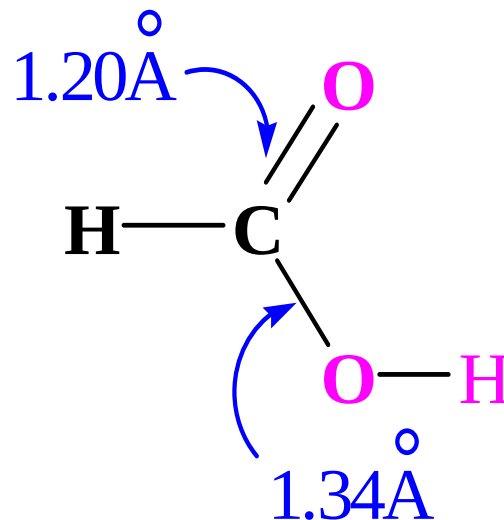
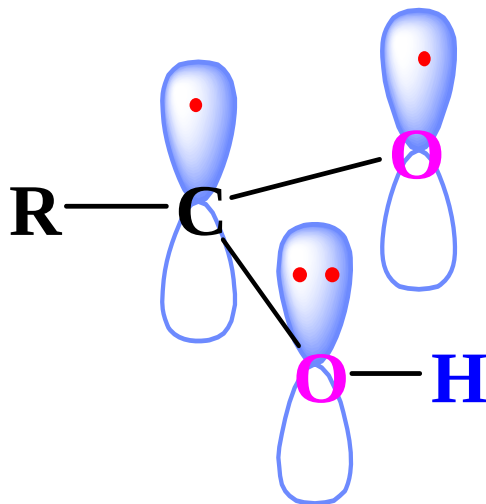
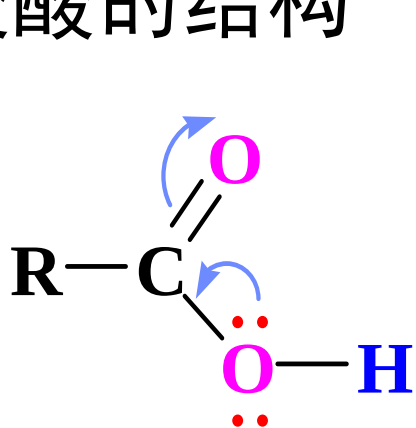
羧酸衍生物的反应

6

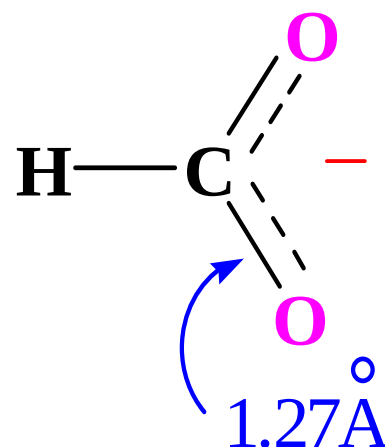
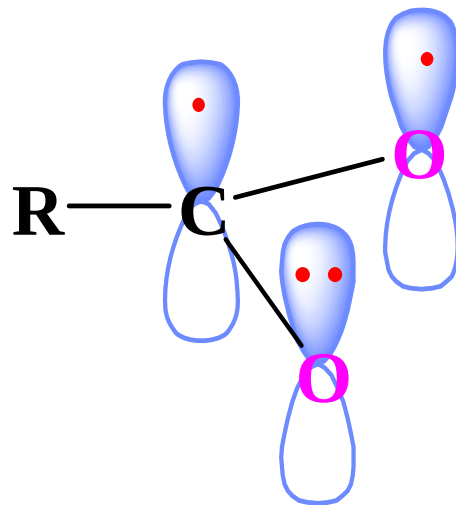
酰胺的特殊性质

12.1 羧酸的结构、命名和物理性质

1. 羧酸的结构



羧酸根的共振结构



12.1 羧酸的结构、命名和物理性质

2. 羧酸的命名

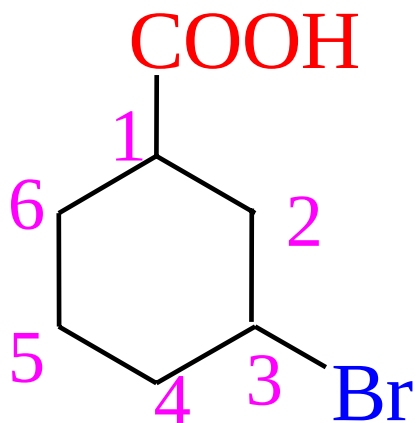
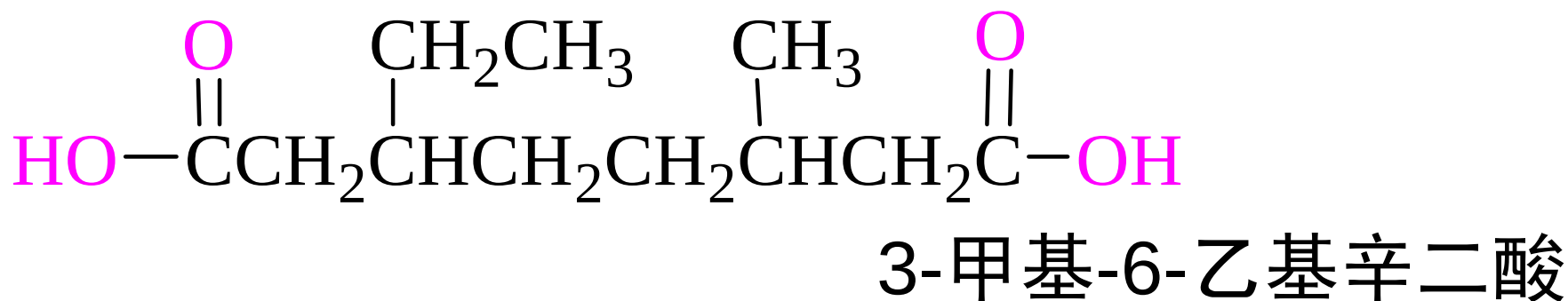
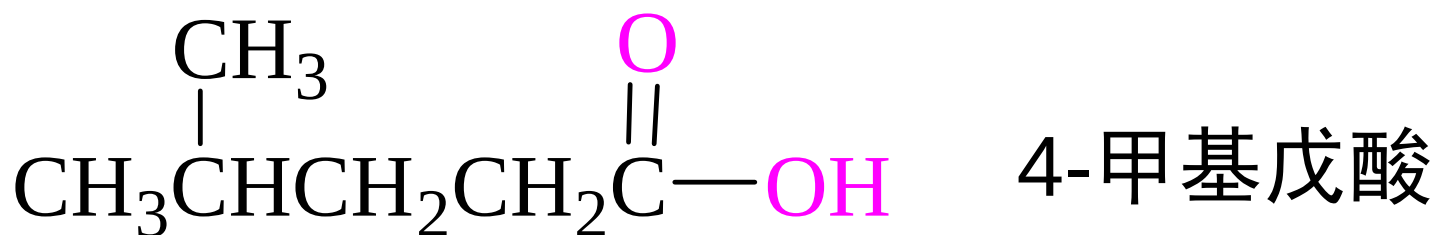
系统命名

- 将相应的烃名改为某酸，羧基始终编为1号
- 主链含烯键→ 某烯酸； 某碳烯酸
- 羧基直接连接在环上的：

烃名 + 羧酸	——	脂肪酸
烃名 + 甲酸	——	芳香酸

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Example

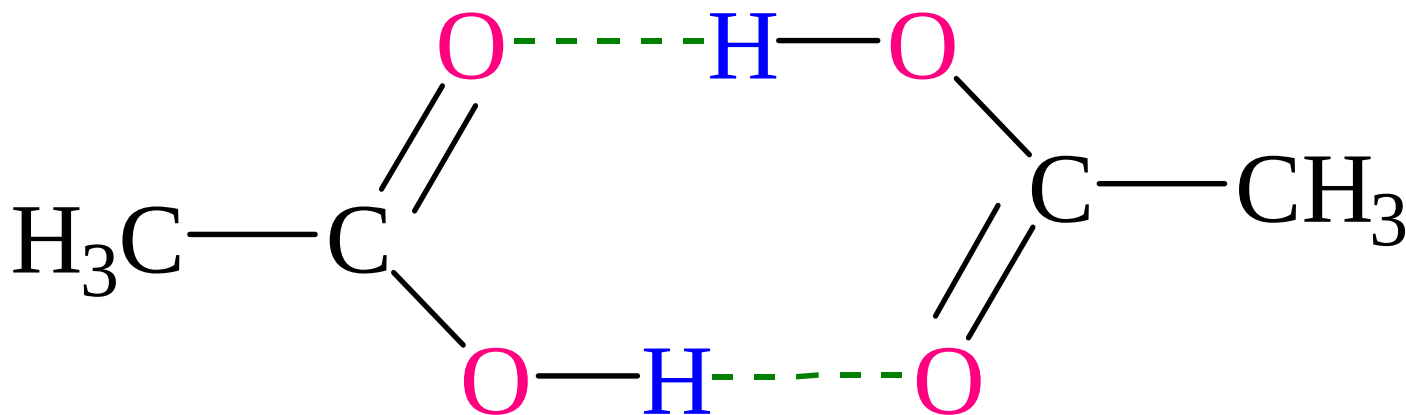


3-溴环己烷羧酸

12.1 羧酸的结构、命名和物理性质

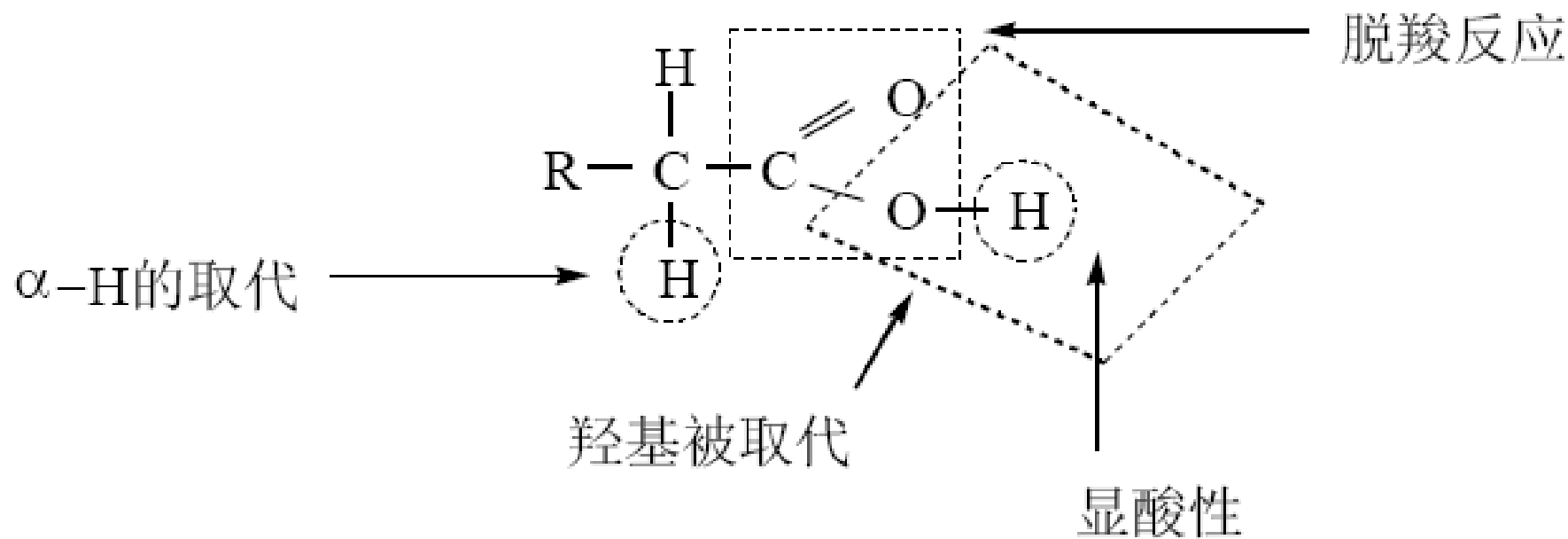
3. 羧酸的物理性质 (自学)

大多数羧酸在固态和液态时以二缔合体形式存在，bp比相应的醇高



乙酸的二缔合体

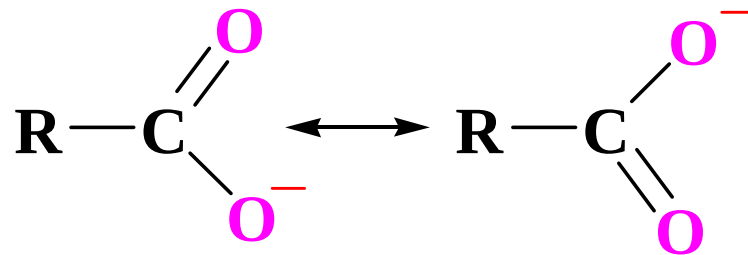
12.2 羧酸的化学性质



12.2 羧酸的化学性质

1. 酸性

- 酸性比醇强得多
- 仍是一种弱酸
- 一元饱和脂肪族羧酸的pK_a值一般在3~5之间



	HCl	CH ₃ COOH	CH ₃ CH ₂ OH
pK _a	-7	4.72	16

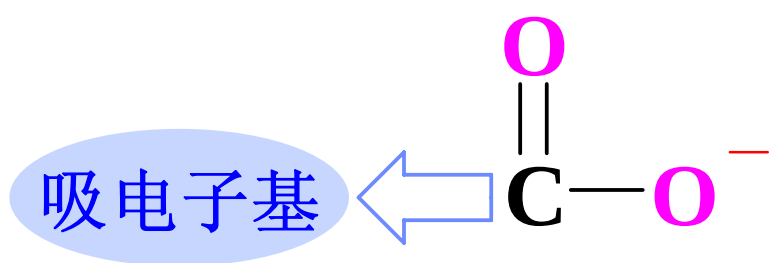


酸性

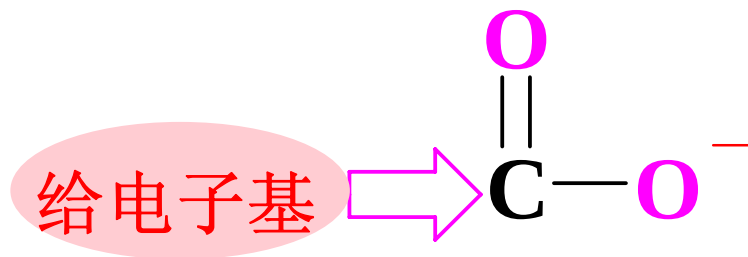
取代基对羧酸酸性的影响

- 吸电子取代基使酸性增强
- 给电子取代基使酸性减弱

Structure	pKa
Cl_2CHCOOH	1.26
ClCH_2COOH	2.85
CH_3COOH	4.72

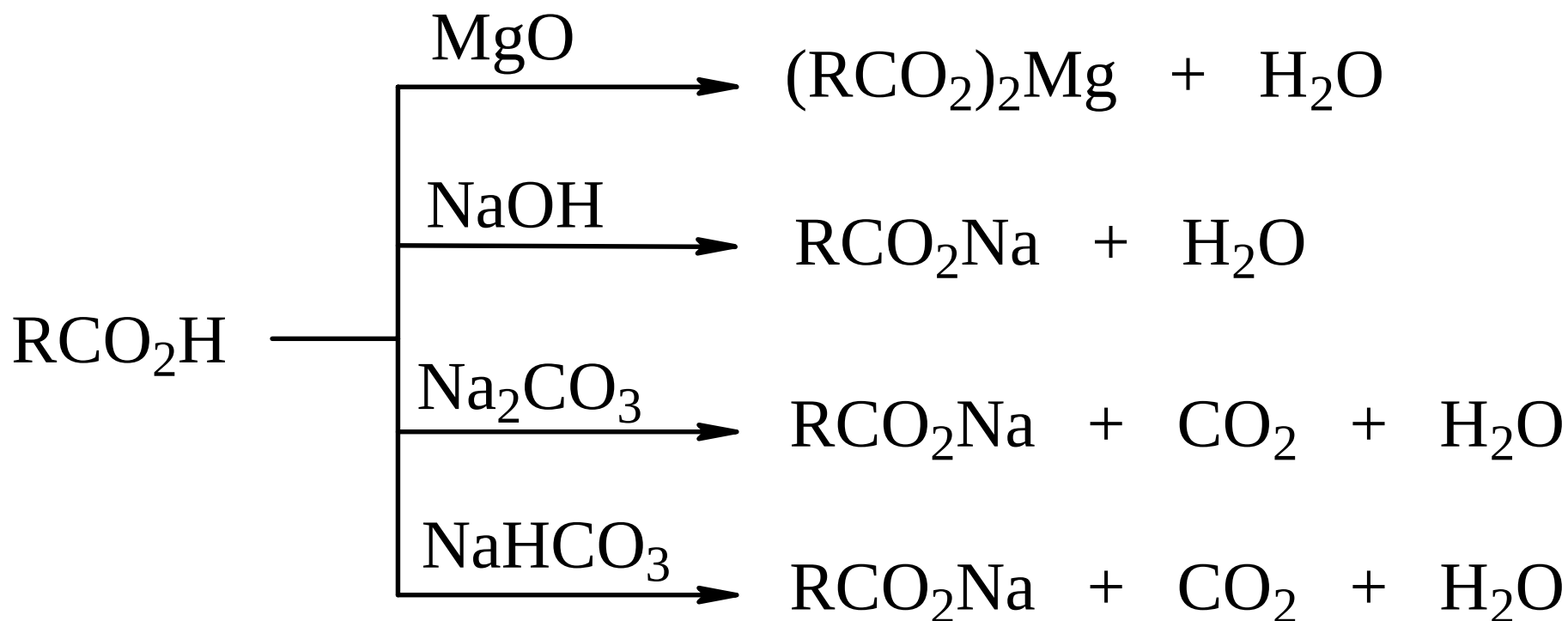


吸电子取代基
提高羧酸盐稳定性



给电子取代基
降低羧酸盐稳定性

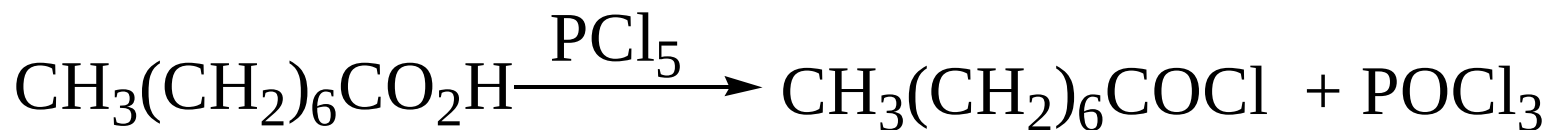
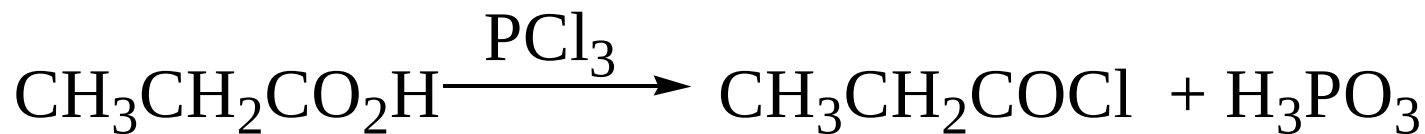
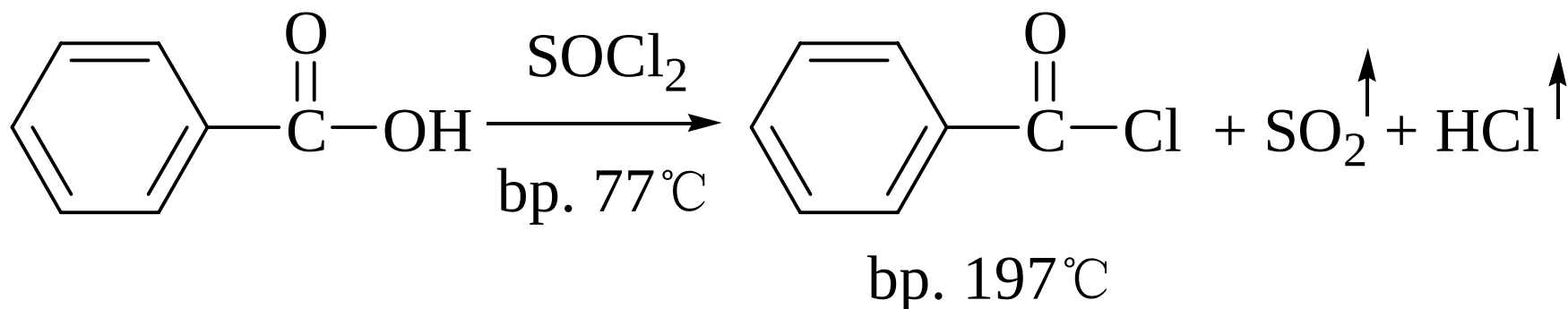
羧酸的酸性反应



12.2 羧酸的化学性质

2. 羧酸衍生物的生成

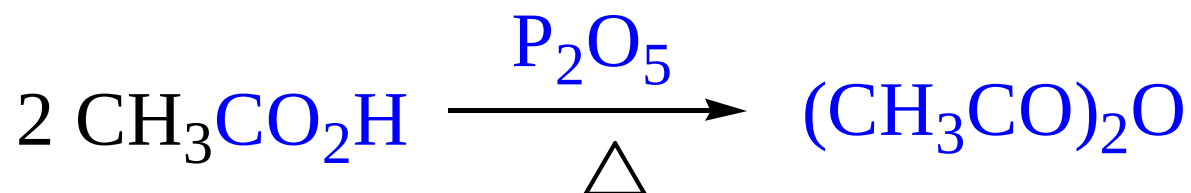
(1) 形成酰卤



12.2 羧酸的化学性质

2. 羧酸衍生物的生成

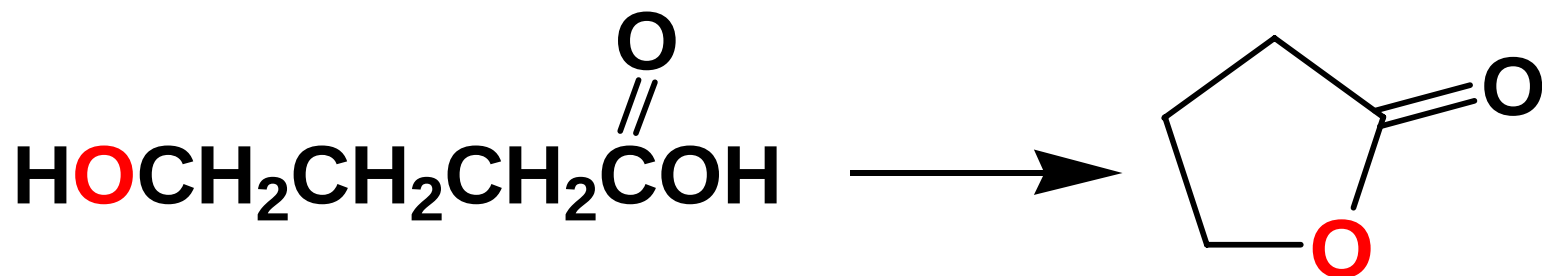
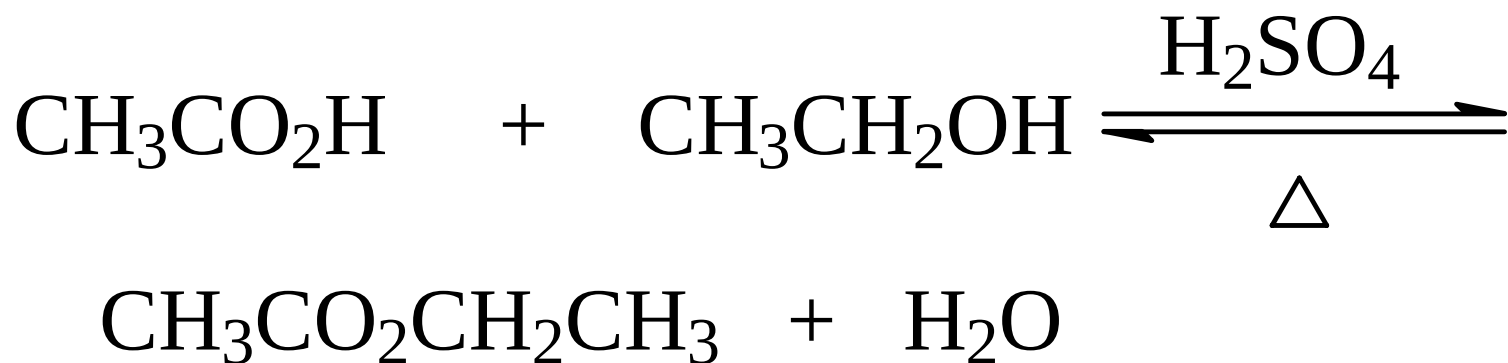
(2) 形成酸酐



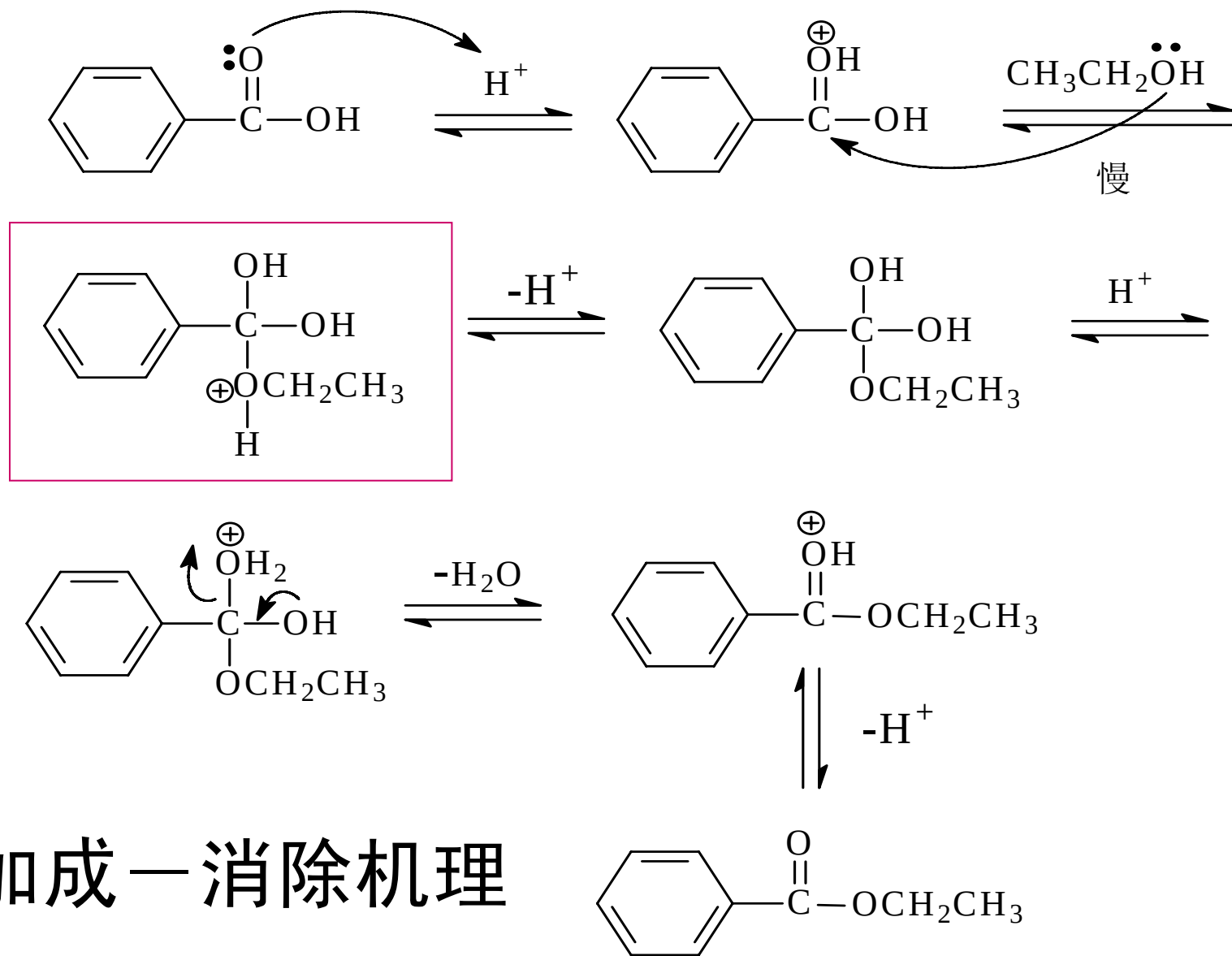
12.2 羧酸的化学性质

2. 羧酸衍生物的生成

(3) 生成酯



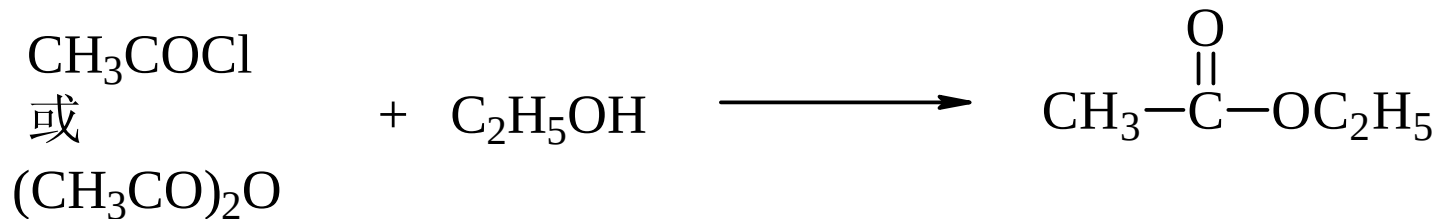
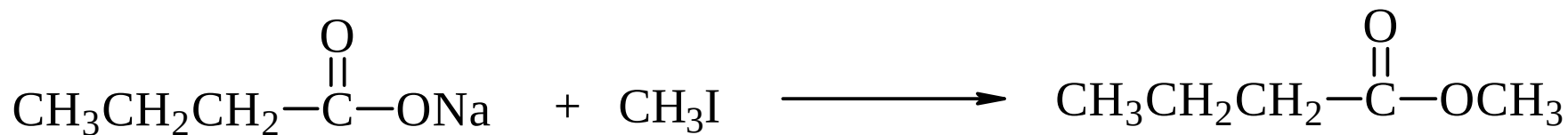
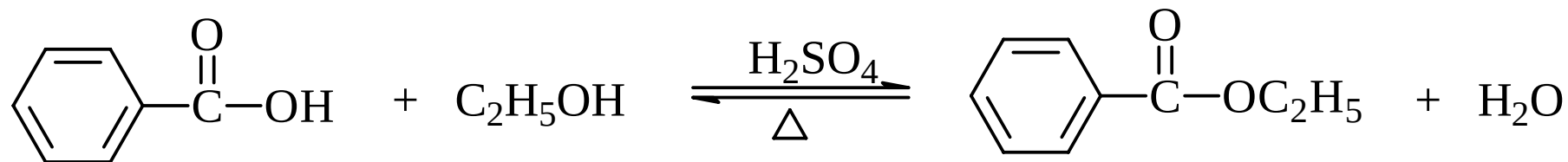
反应机理 (掌握)



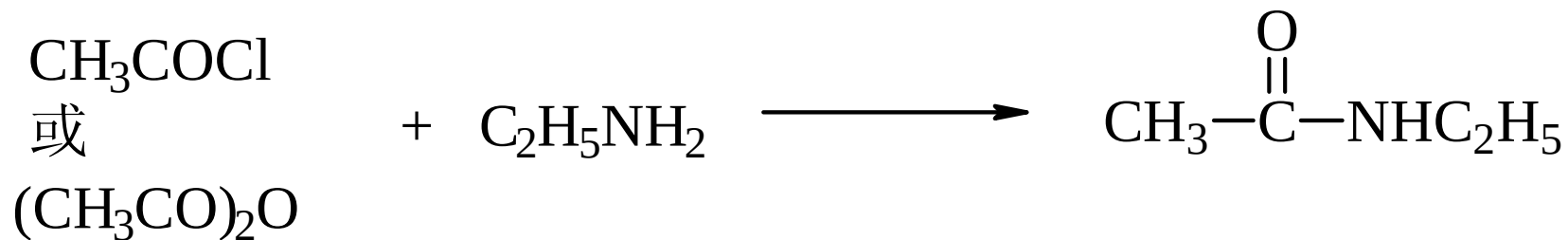
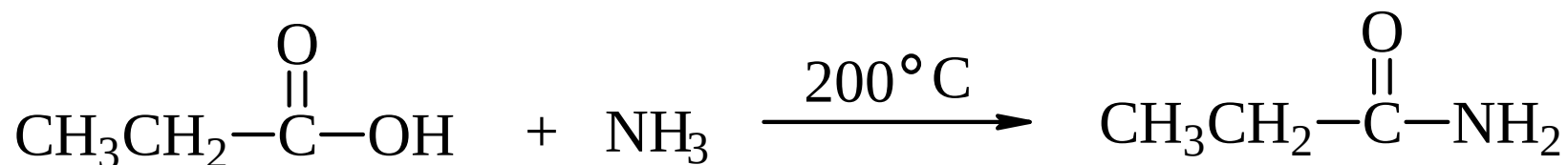
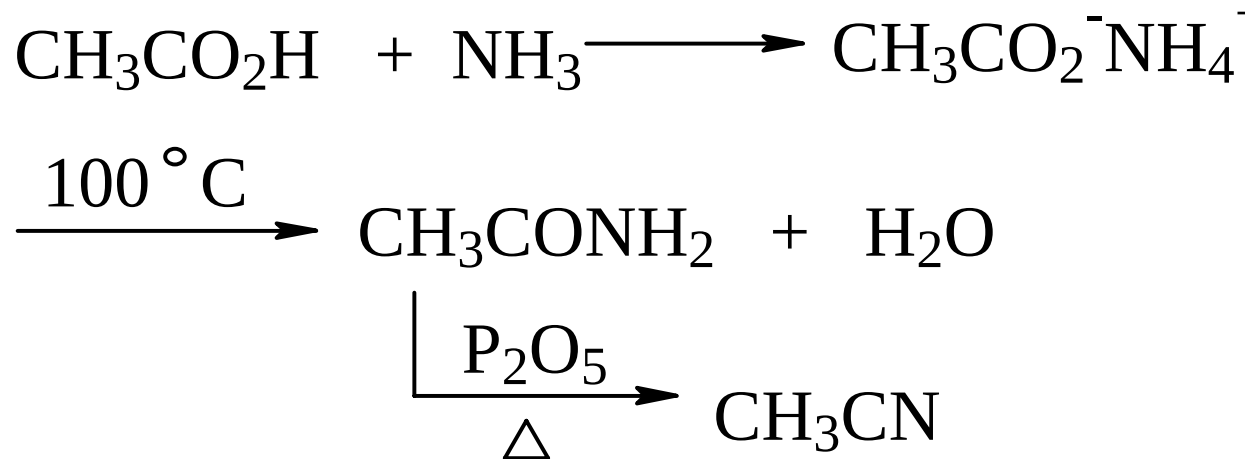
加成—消除机理

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Example

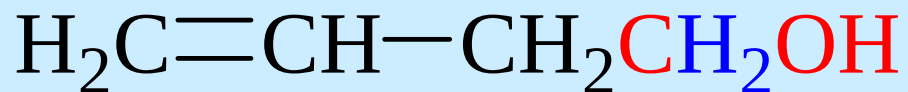
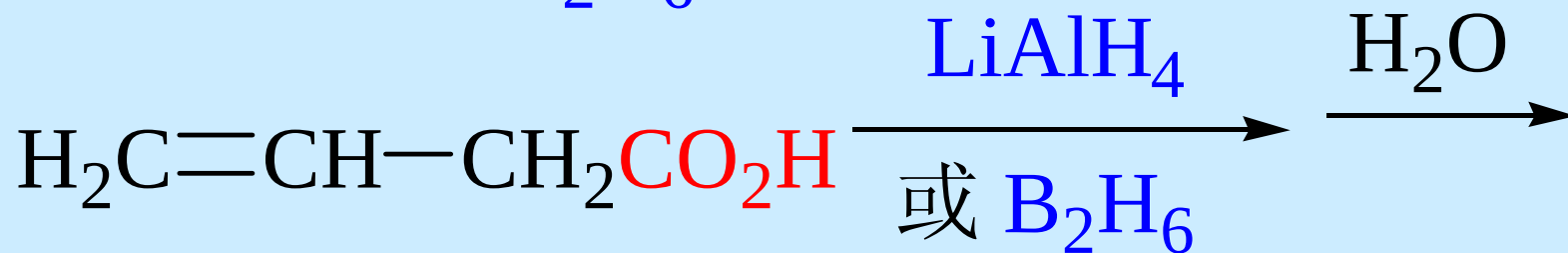
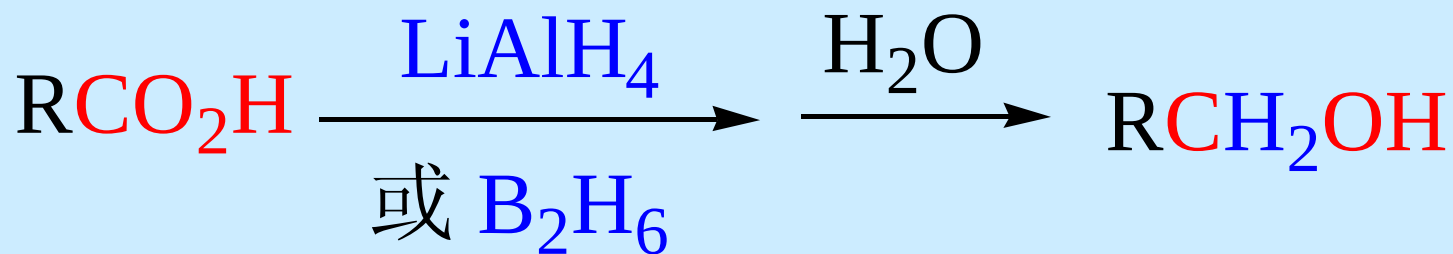


(4) 生成酰胺和腈



12.2 羧酸的化学性质

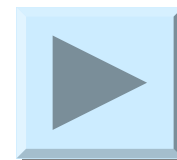
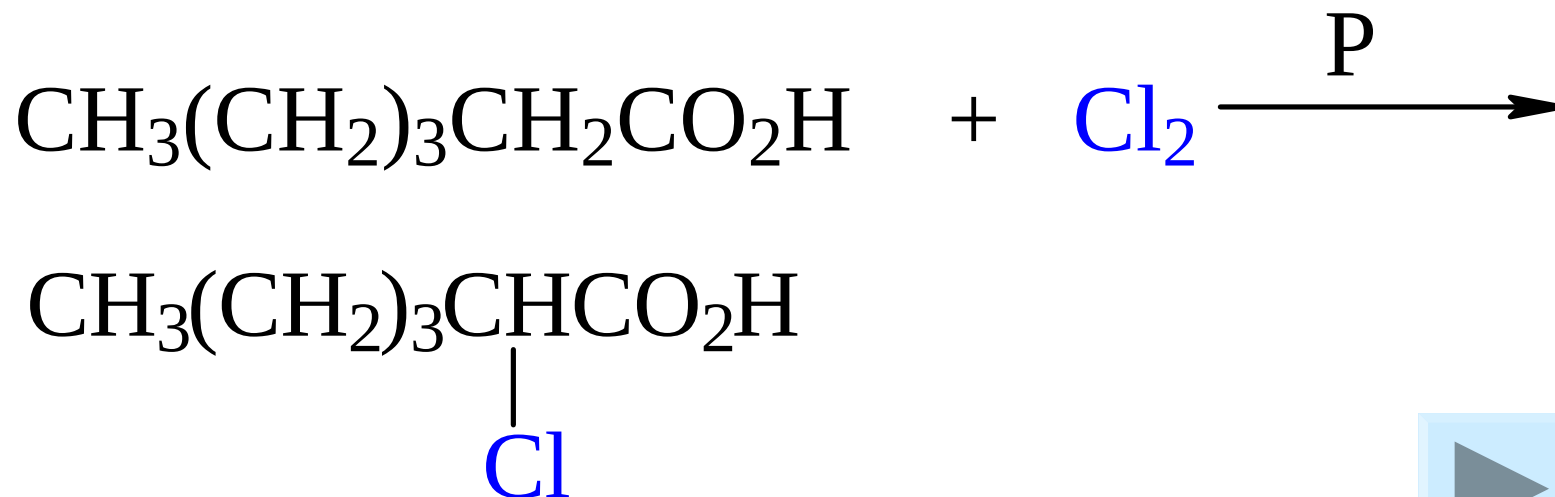
3. 羧酸的还原



12.2 羧酸的化学性质

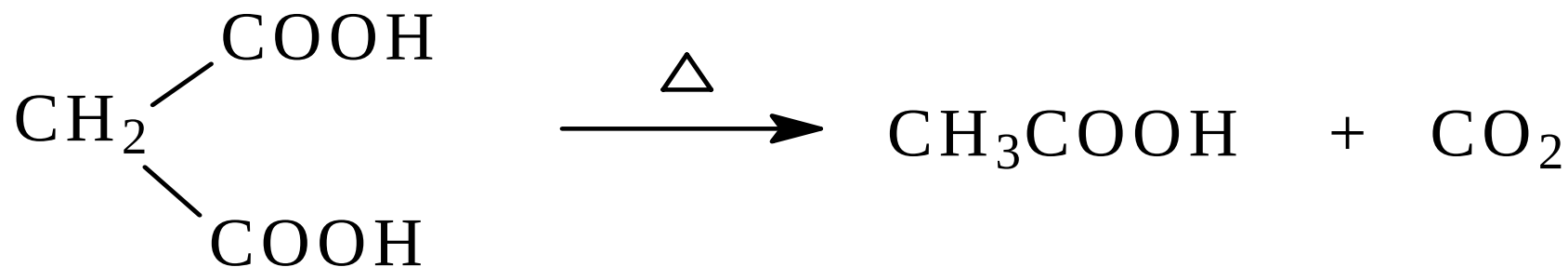
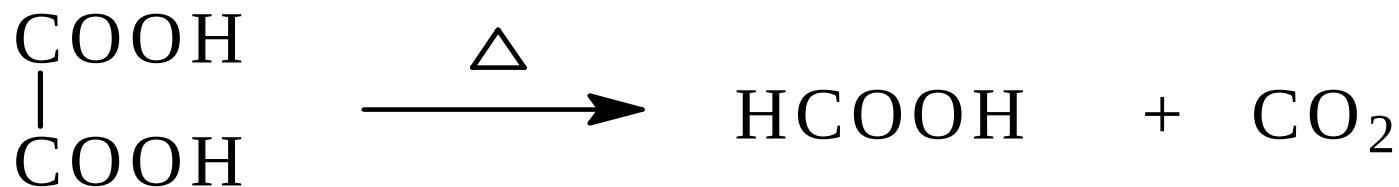
4. 羧酸 α -卤代反应

与卤素在红磷或酰卤催化下发生 α -卤代
——本质上是酰卤的 α -卤代

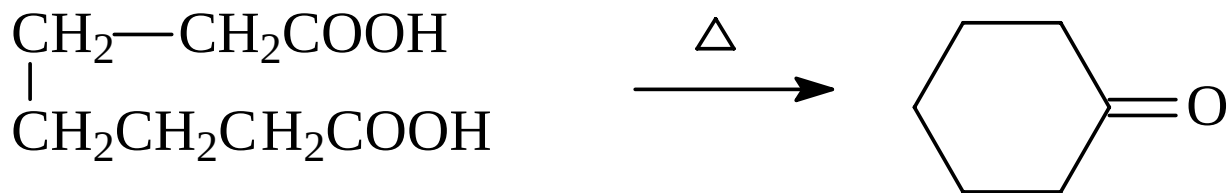
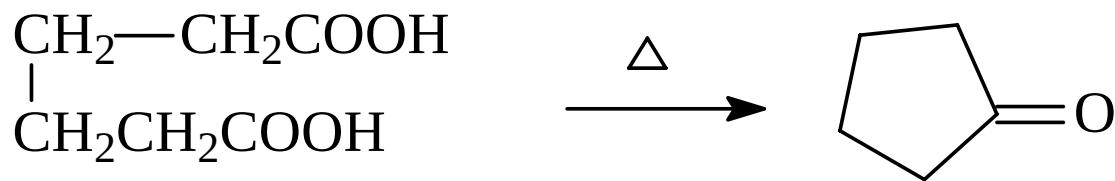
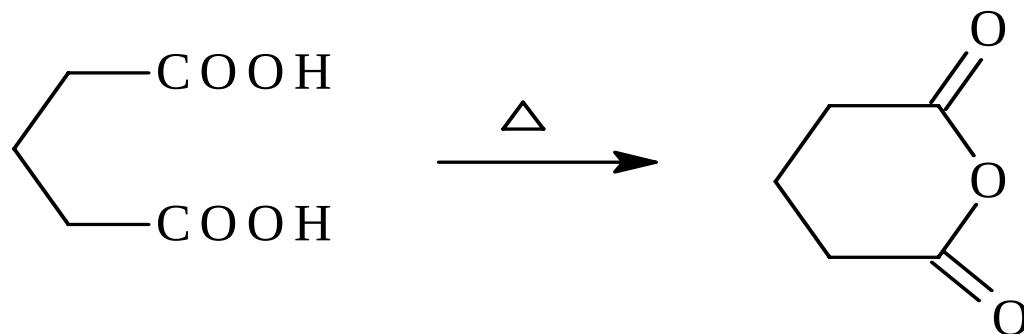
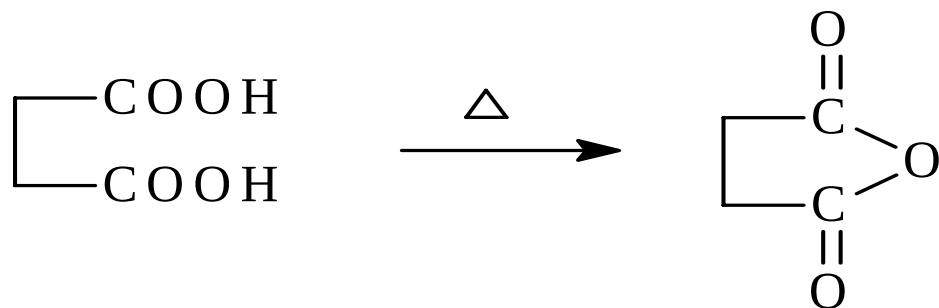


12.2 羧酸的化学性质

5. 二元羧酸受热的反应



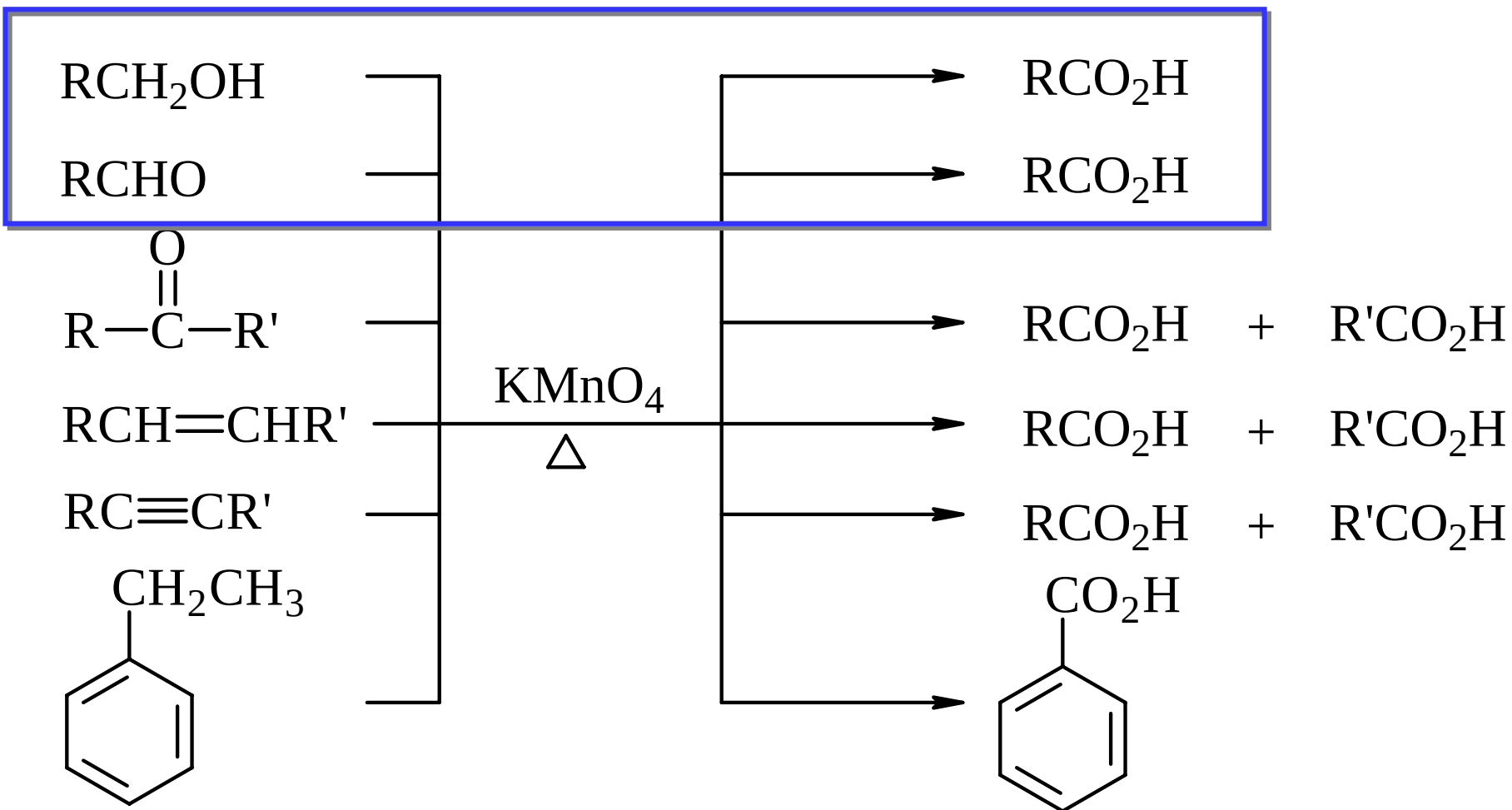
如果有成环的可能时，一般总是形成五员和六员环



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12.3 羧酸的制备

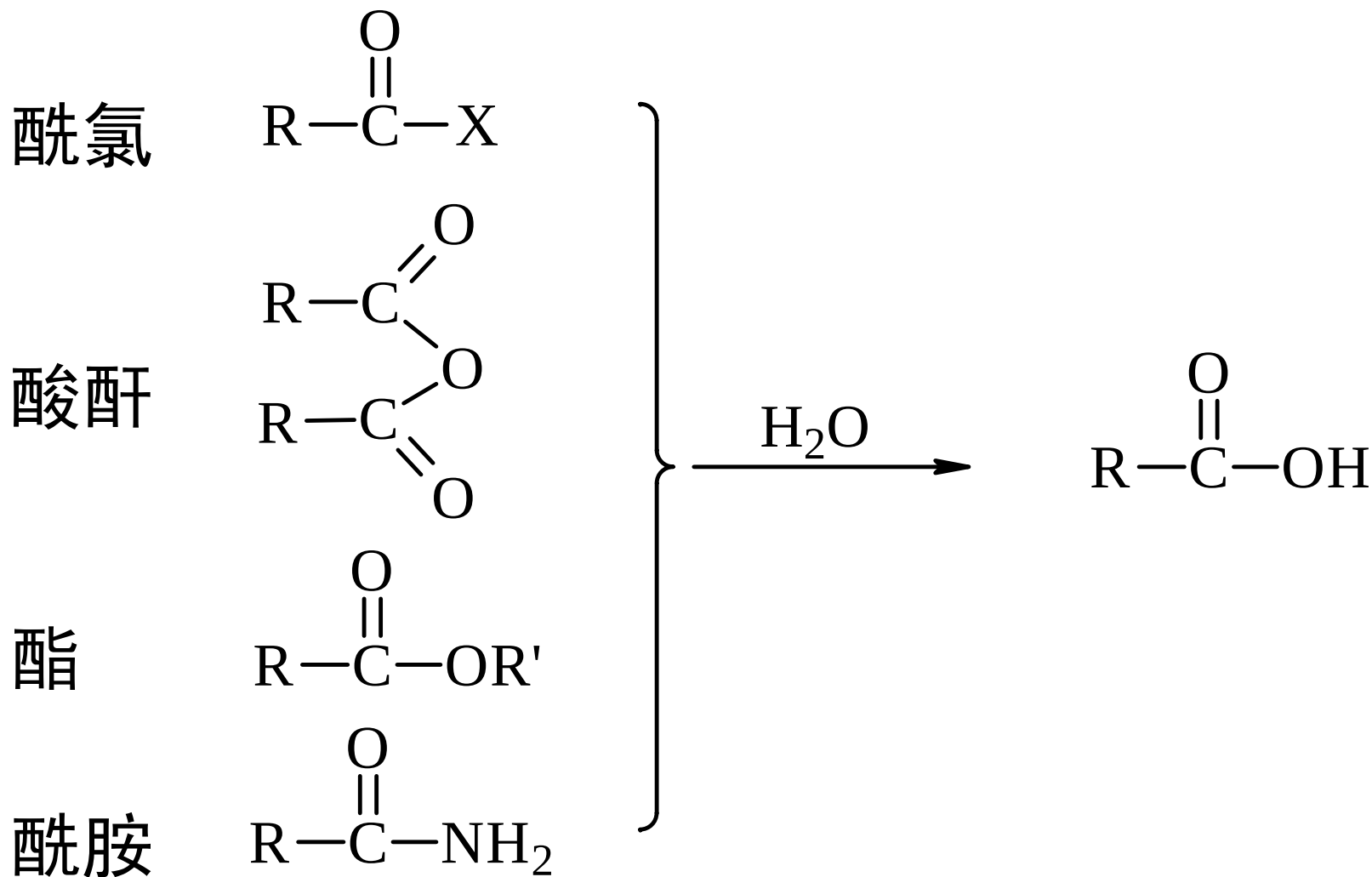
1. 通过氧化反应制备

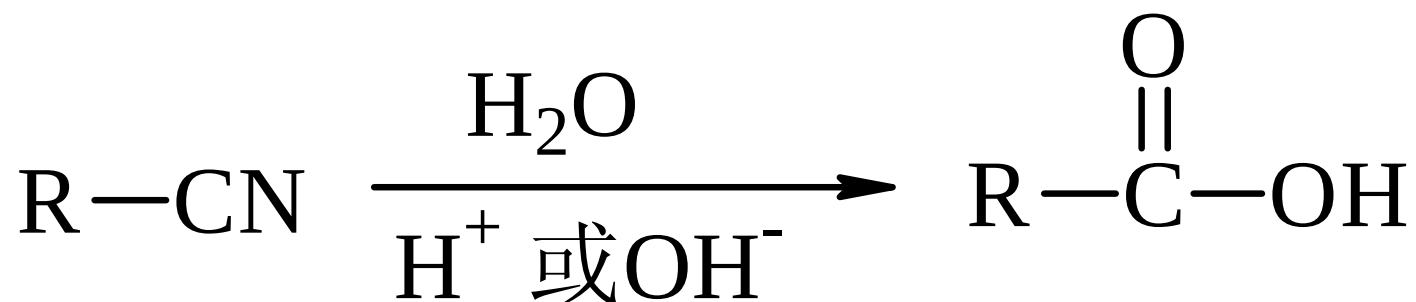


厦门大学有机化学B

12.3 羧酸的制备

2. 由羧酸衍生物水解

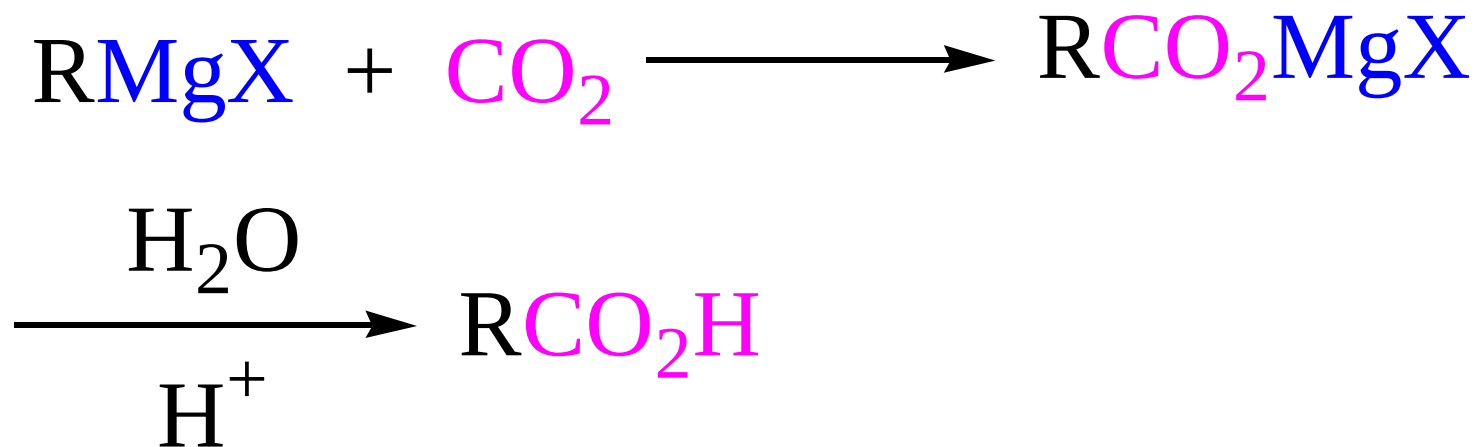




可由卤代烃制备增加一个碳原子的羧酸

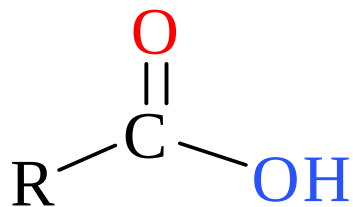
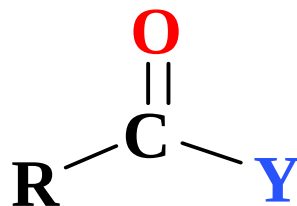
12.3 羧酸的制备

3. 格式试剂与二氧化碳反应制备

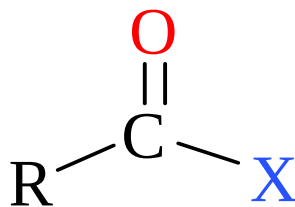


12.4 羧酸衍生物命名和物理性质

羧酸衍生物

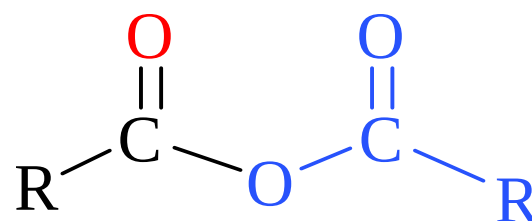


羧酸

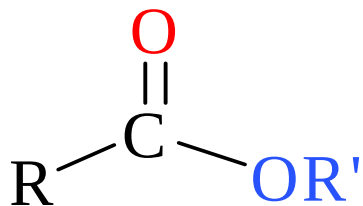


$\text{X} = \text{F}, \text{Cl}, \text{Br}, \text{I}$

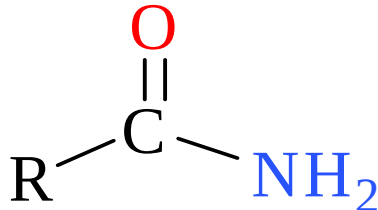
酰卤



酸酐



酯



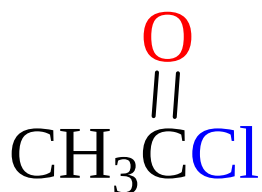
酰胺



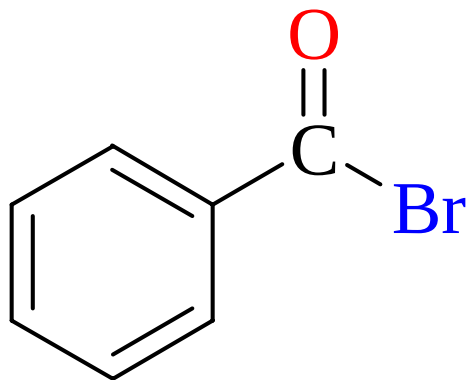
腈

酰卤的命名: RCOX

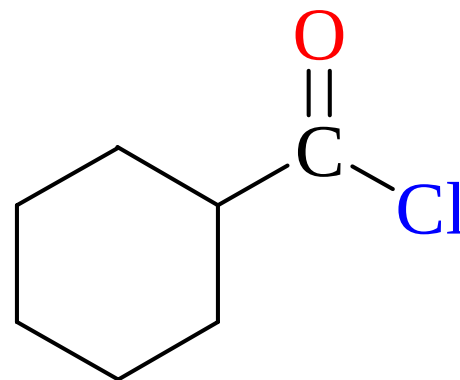
酰基名 + 卤素名



乙酰氯



苯甲酰氯

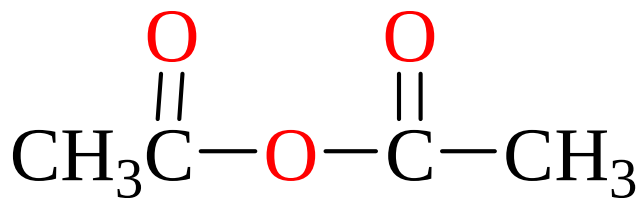


环己烷甲酰氯

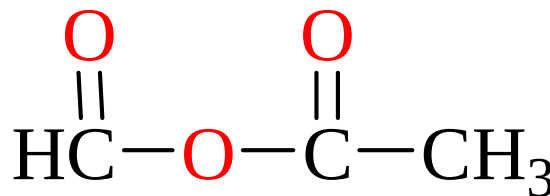
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酸酐的命名: $\text{RCO}_2\text{COR}'$

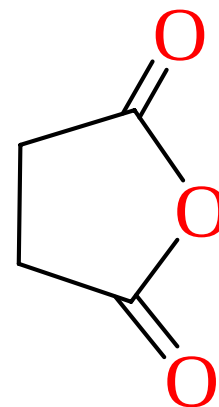
× 酸 → × (酸) 酐; × 酸 × 酸酐



乙酸酐

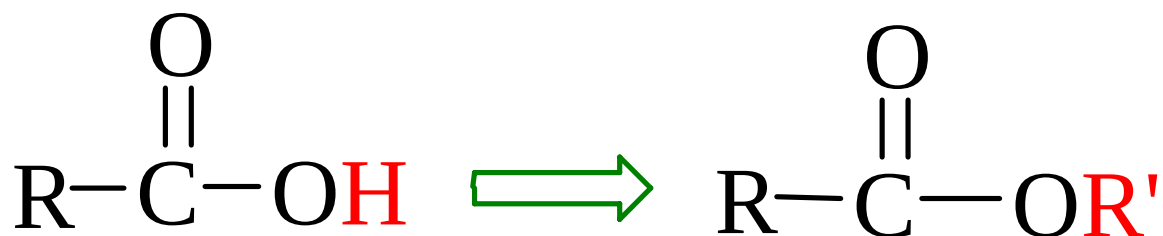


甲(酸)乙(酸)酐



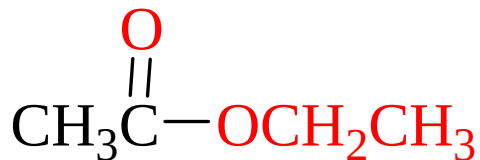
丁二酸酐

酯的命名: $\text{RCO}_2\text{R}'$

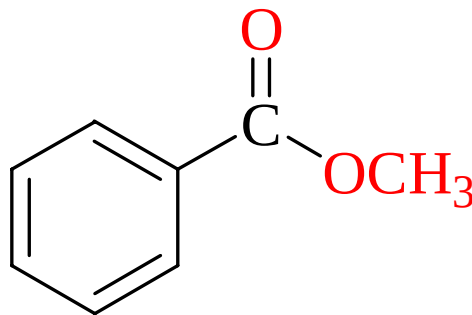


羧酸的氢被烃基取代

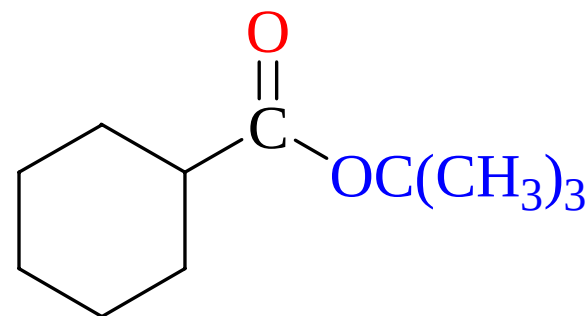
× 酸 × (烃基名) 酯



乙酸乙酯



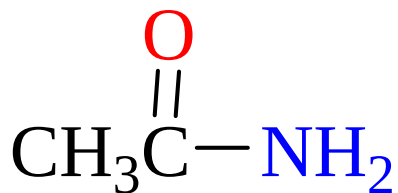
苯甲酸甲酯



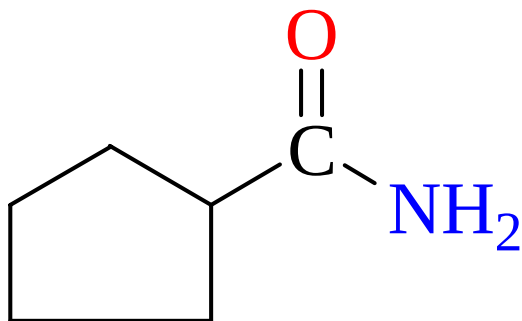
环己烷羧酸叔丁酯

酰胺的命名: RCONH_2

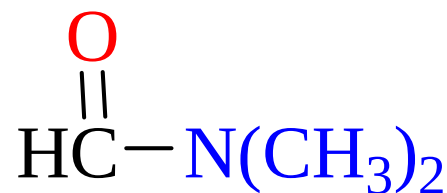
× 酸 → × 酰胺 N-烃基 × 酰胺



乙酰胺



环戊烷羧酰胺

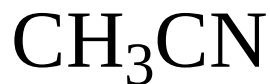


N,N-二甲基甲酰胺

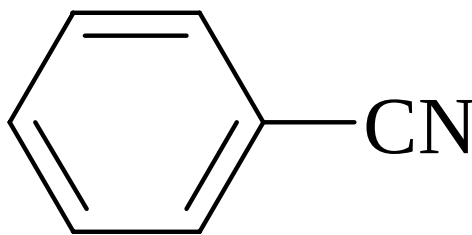
DMF

腈的命名: RCN

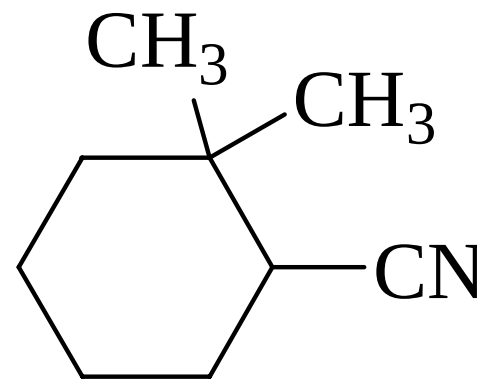
腈 × 腈(含CN碳原子)



乙腈



苯甲腈

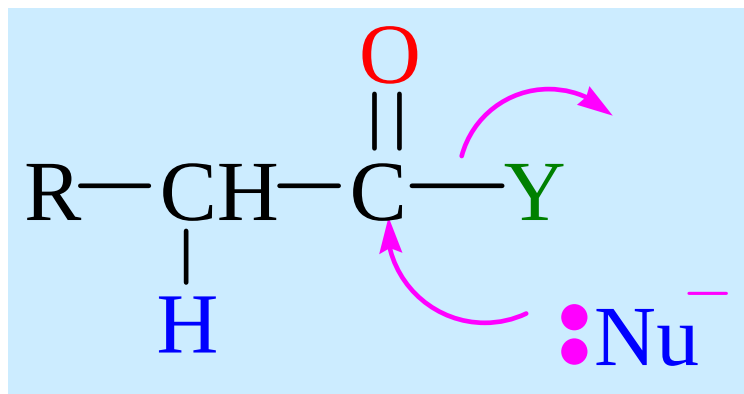


2,2-二甲基环己烷甲腈

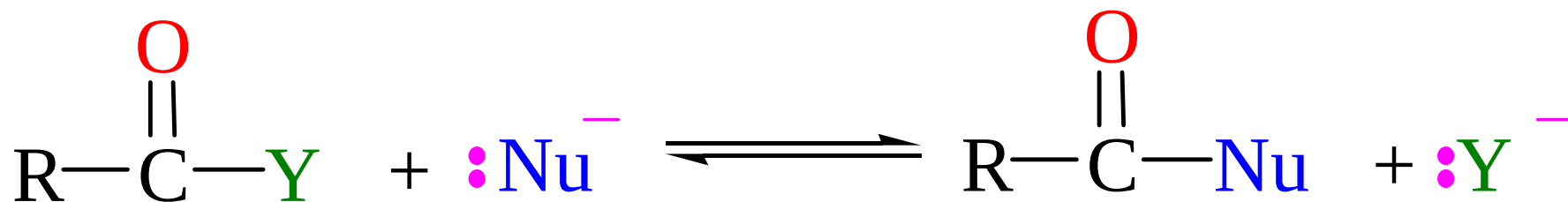
羧酸衍生物物理性质 (自学)

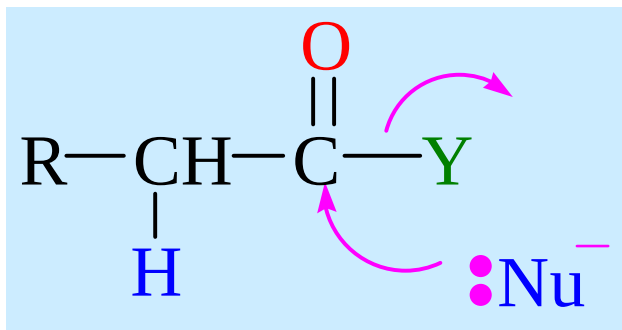
12.5 羧酸衍生物的反应

12.5.1 亲核取代反应

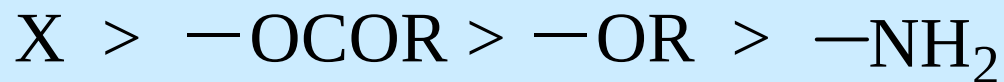


机理——加成-消除机制

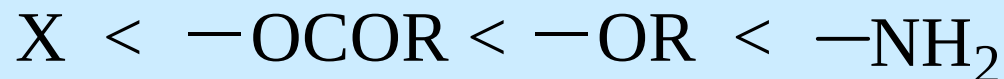




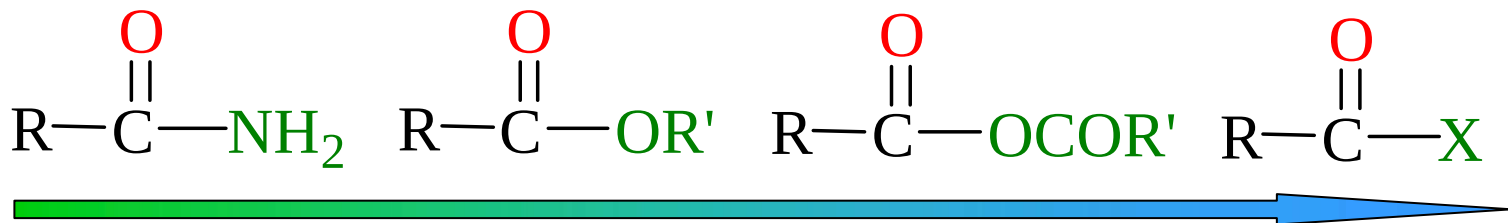
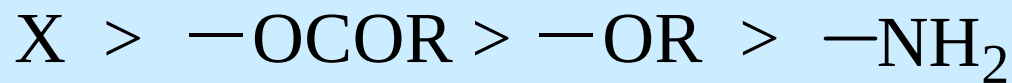
诱导效应



共轭效应

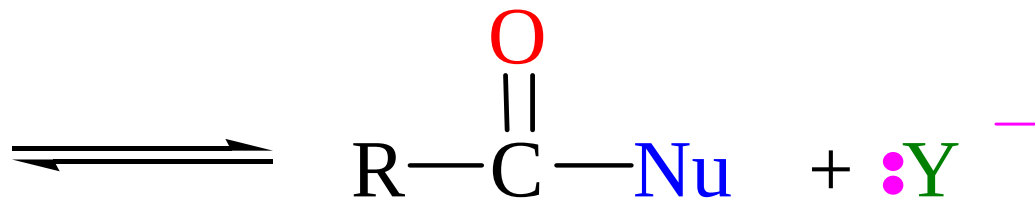
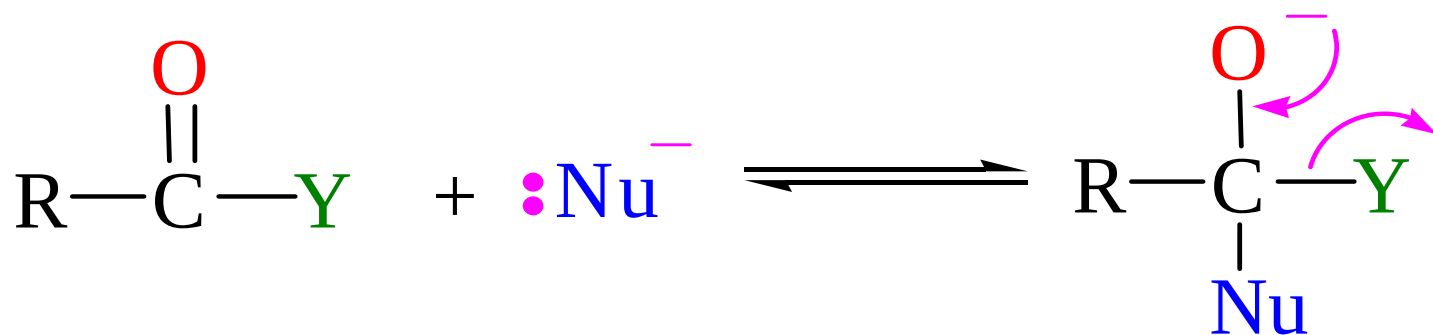
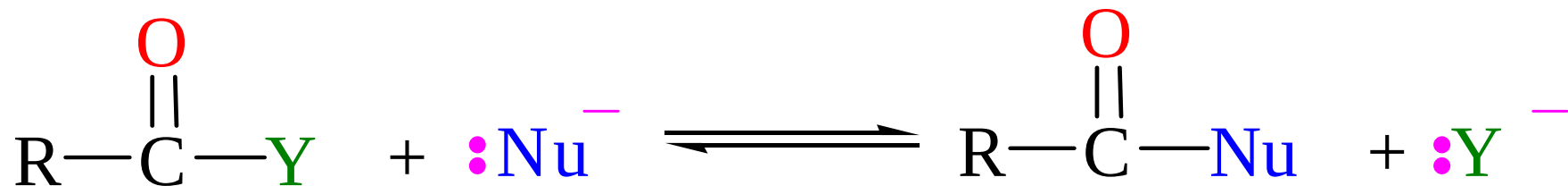


离去能力

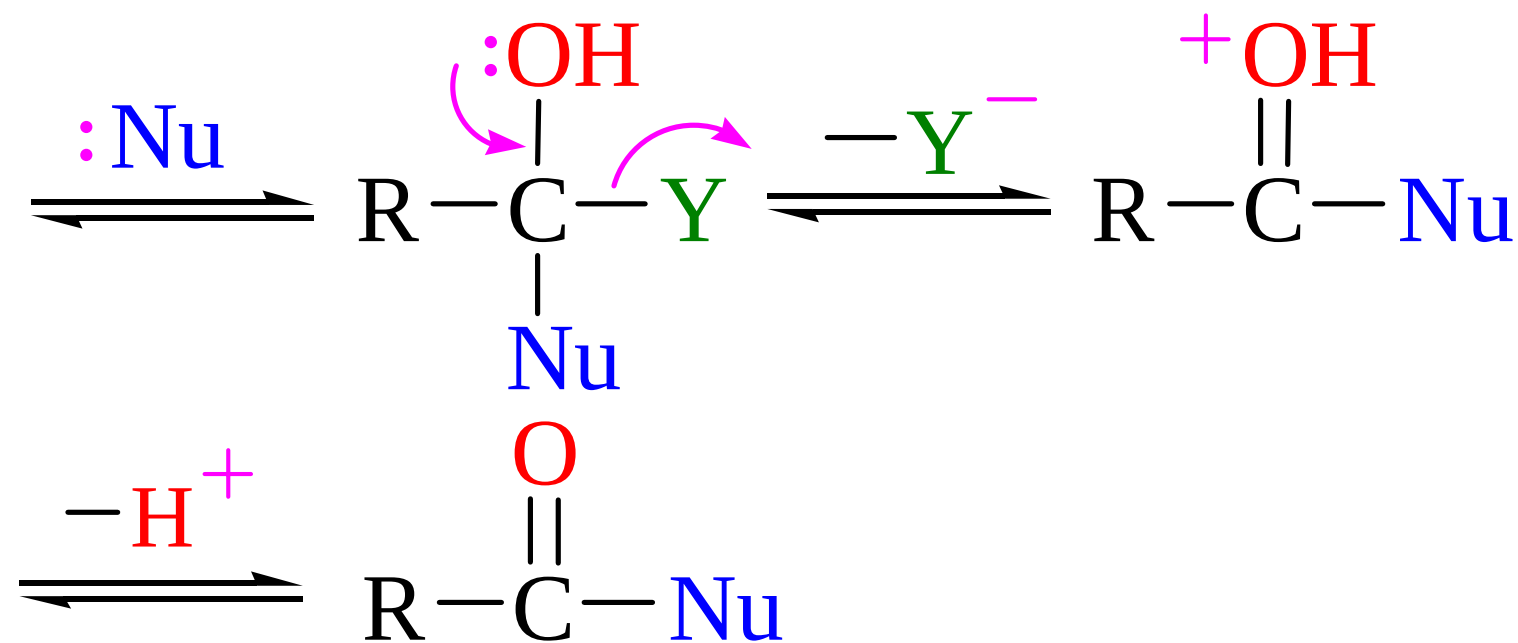
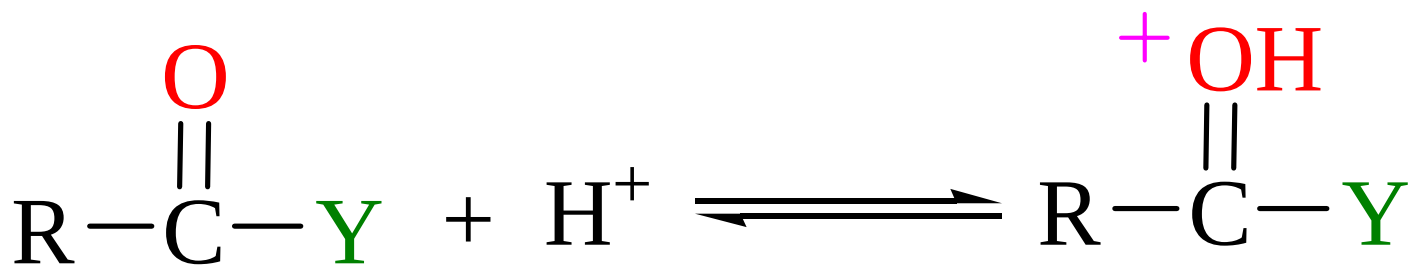


反应性增强

碱催化下的反应机理



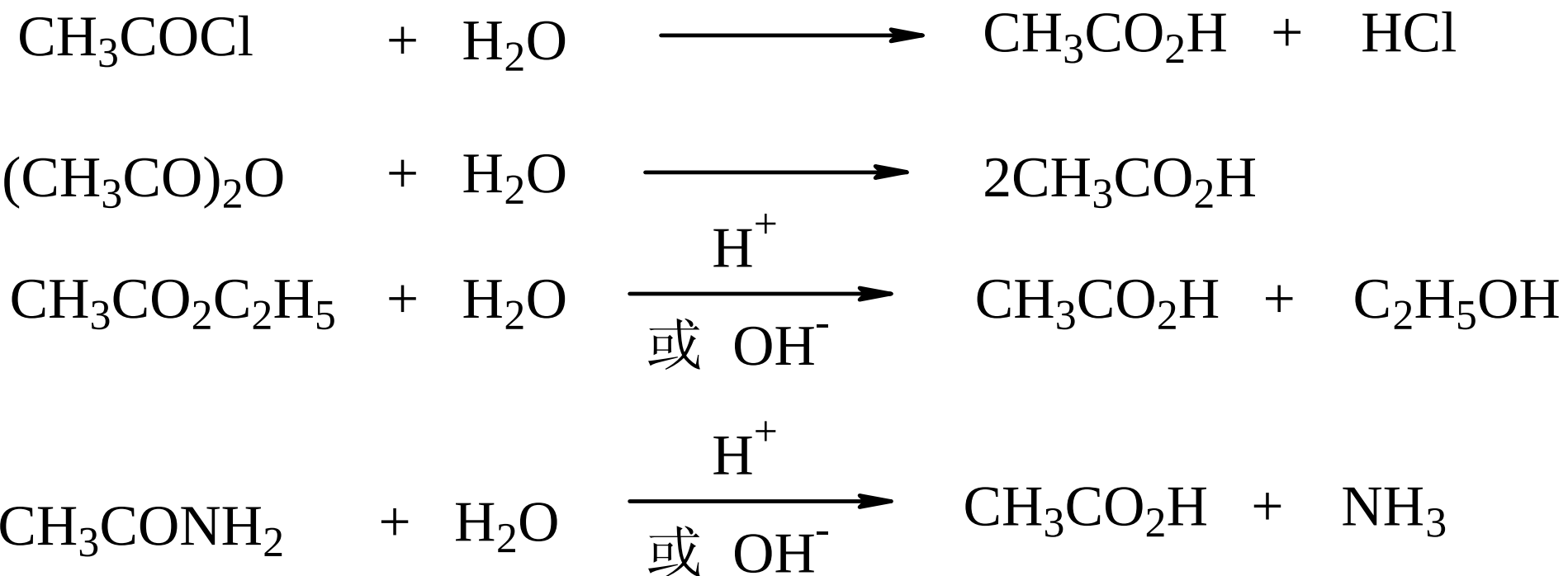
酸催化下的反应机理



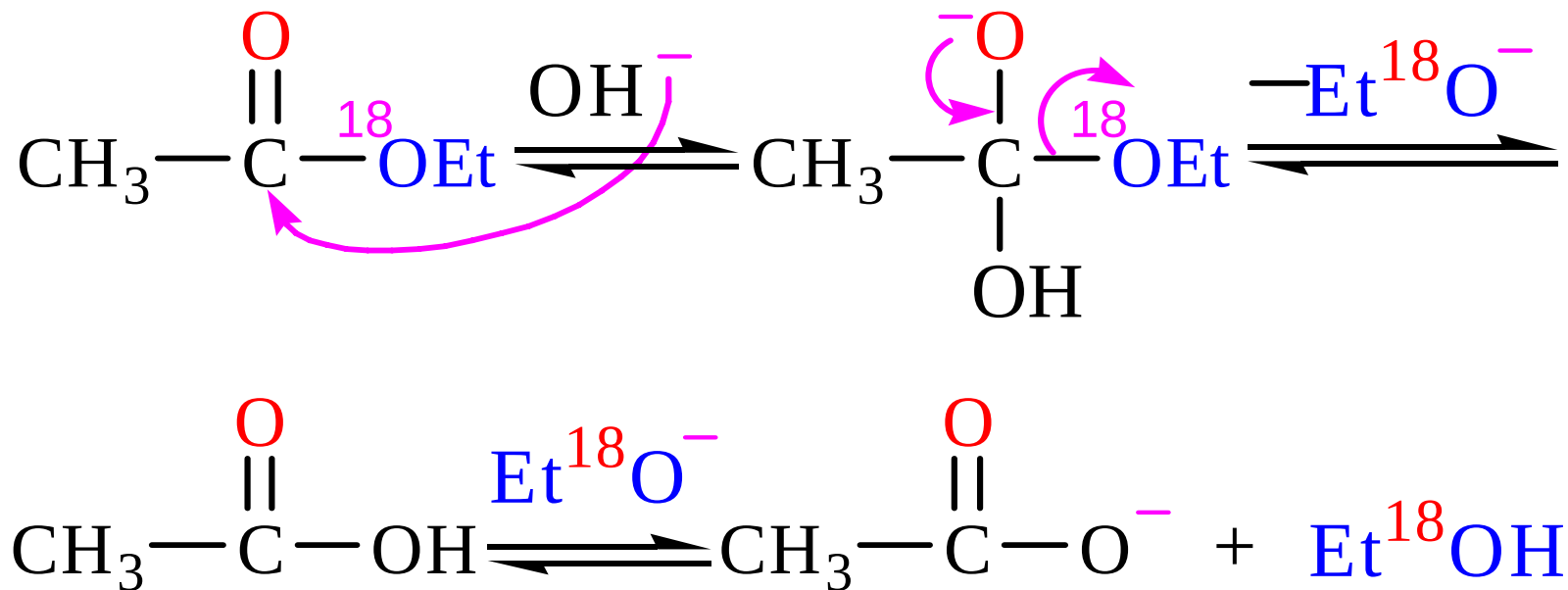
厦门大学有机化学B

12.5.1 亲核取代反应

1. 水解反应

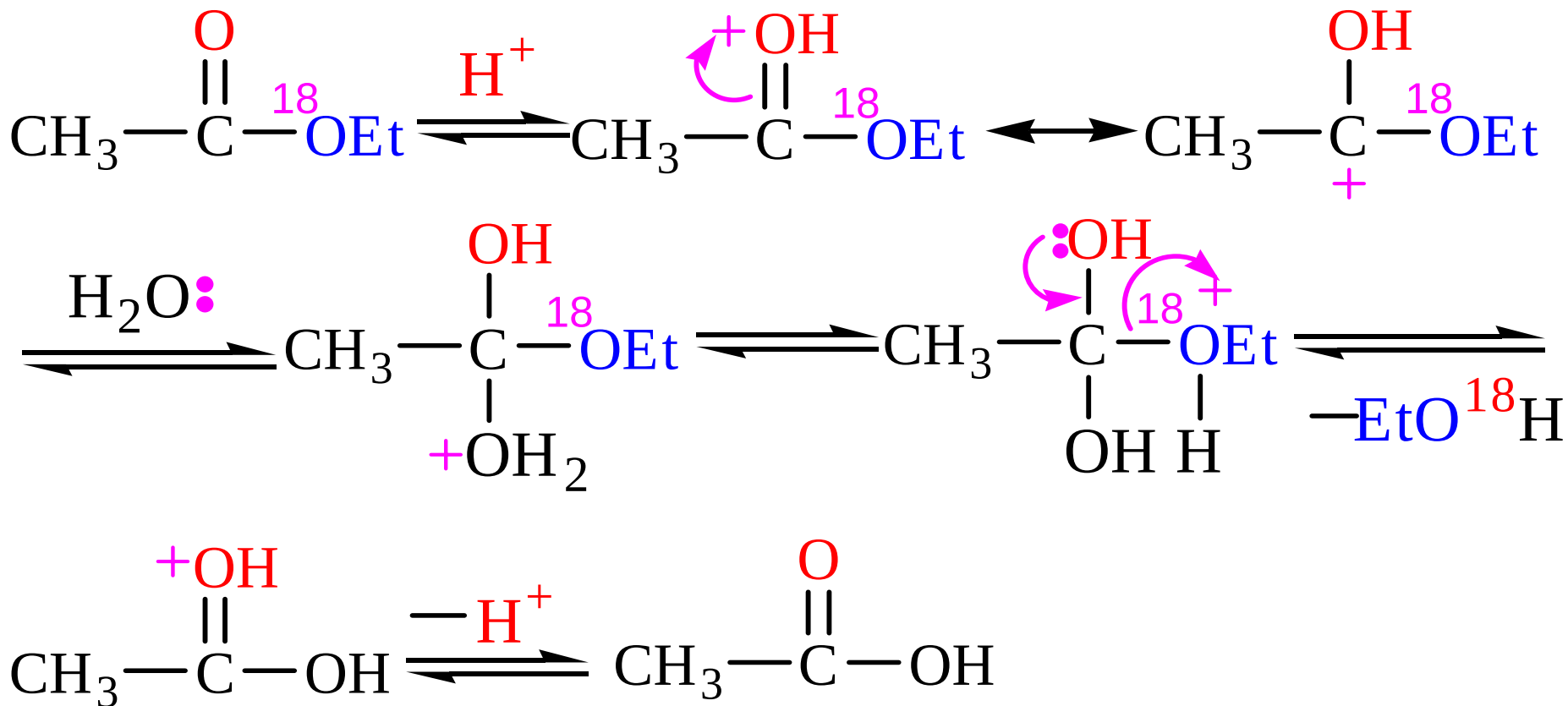


碱催化下酯的水解反应机理

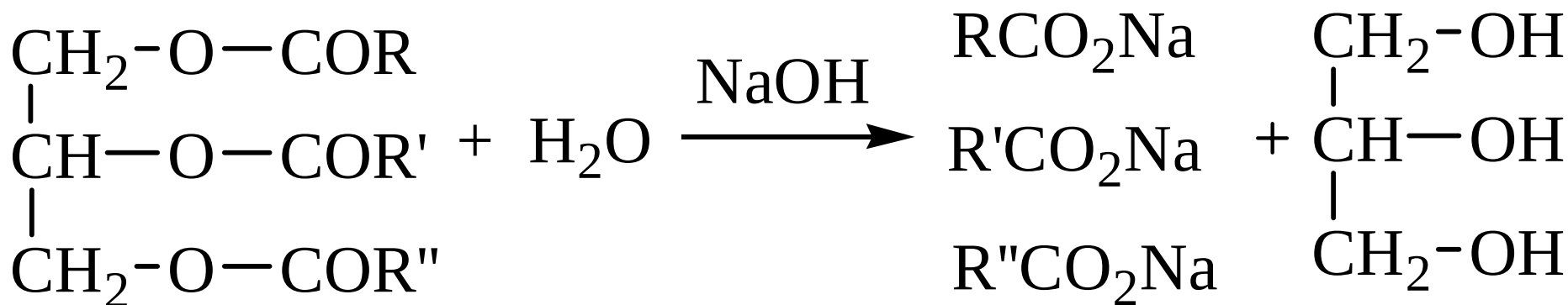


经同位素证明酯的水解是酰氧断裂
而不是烷氧断裂

酸催化下酯的水解反应机理



天然油脂的皂化反应

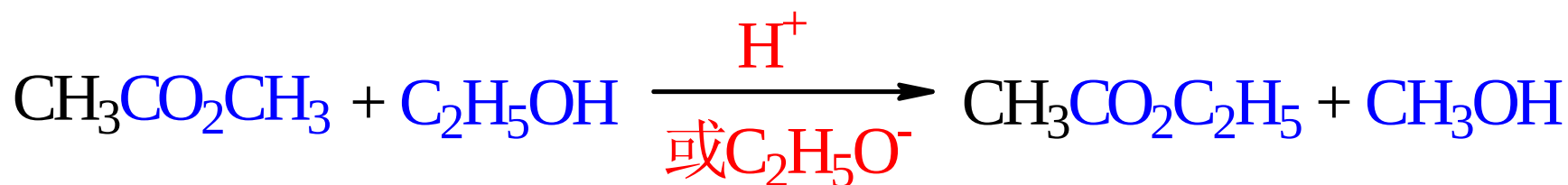


甘油三酯

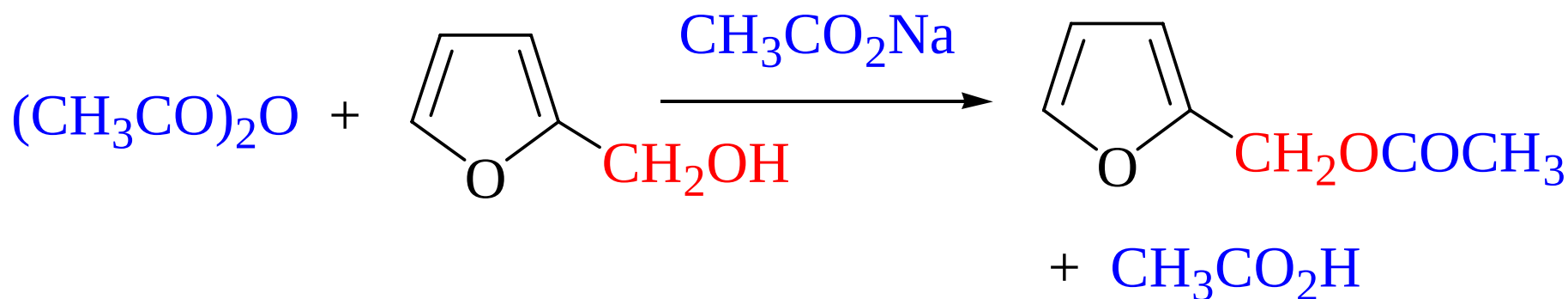
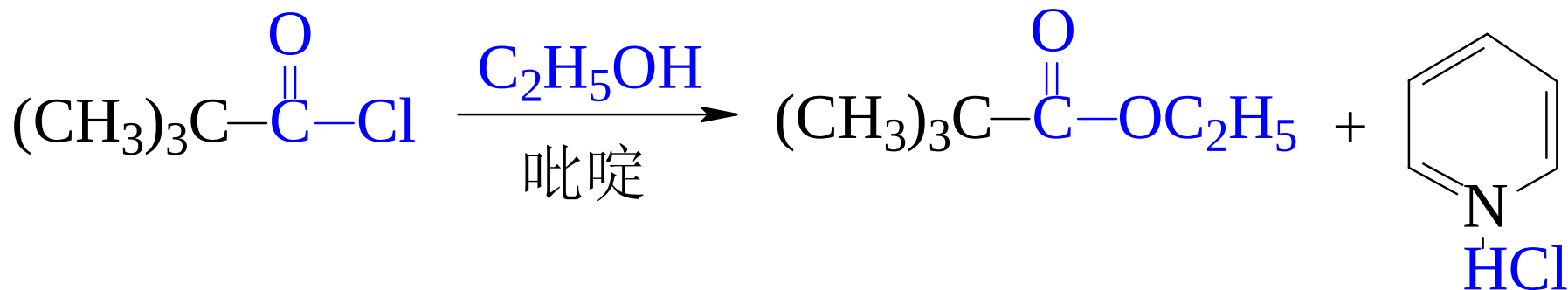
甘油醇

12.5.1 亲核取代反应

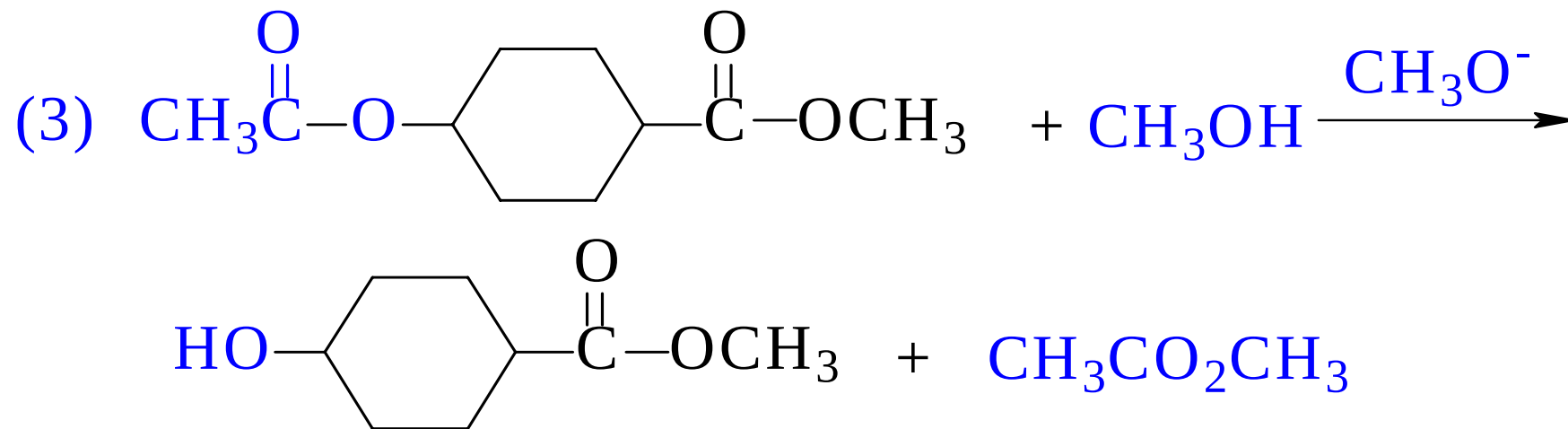
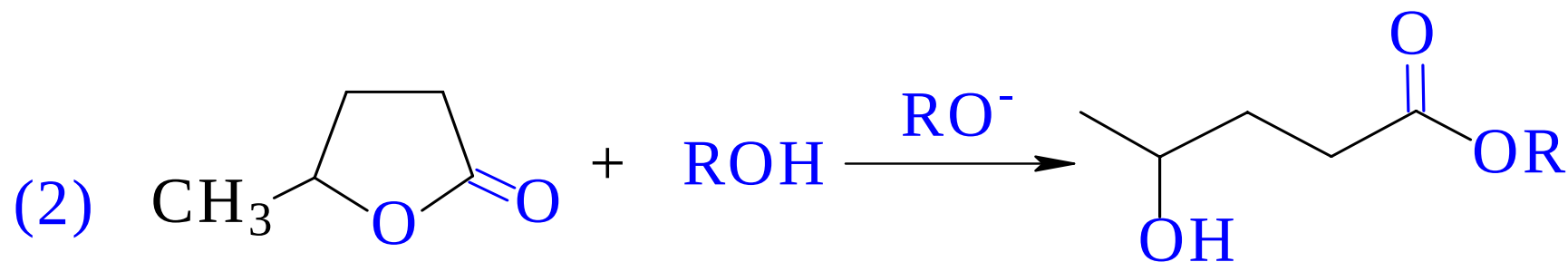
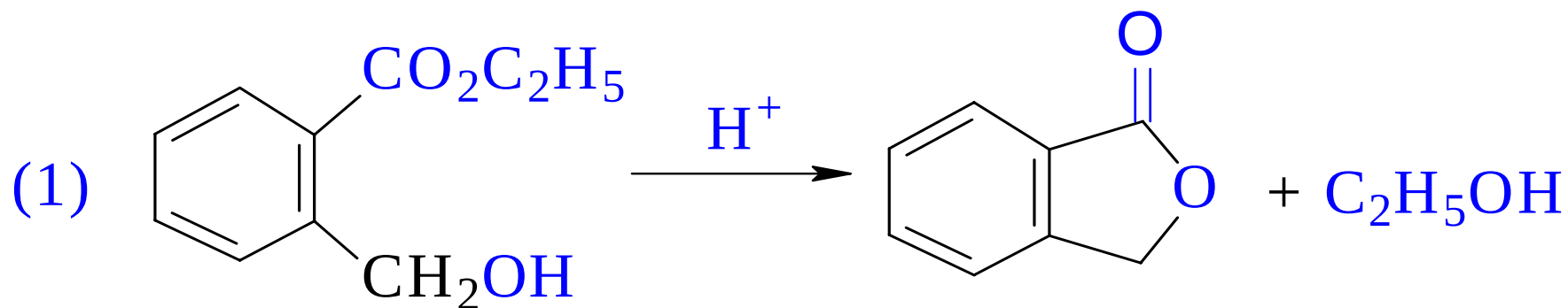
2. 醇解反应



醇解反应实例



酯交换反应的实例

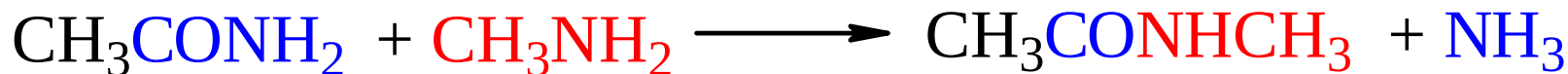
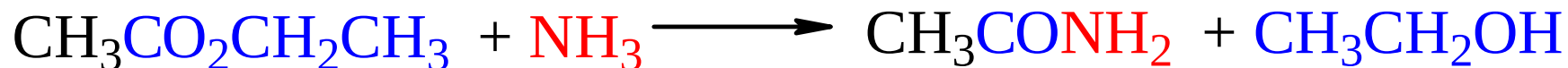
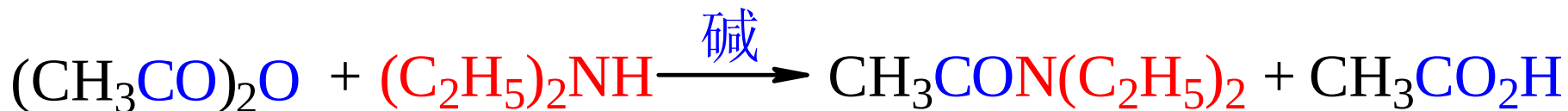
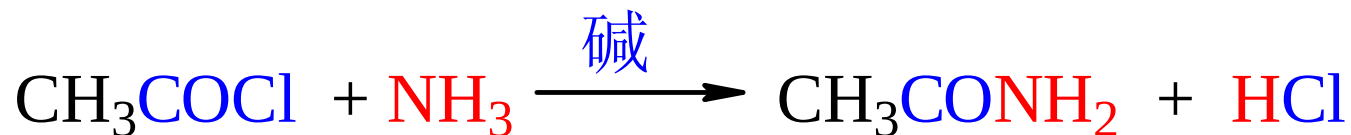


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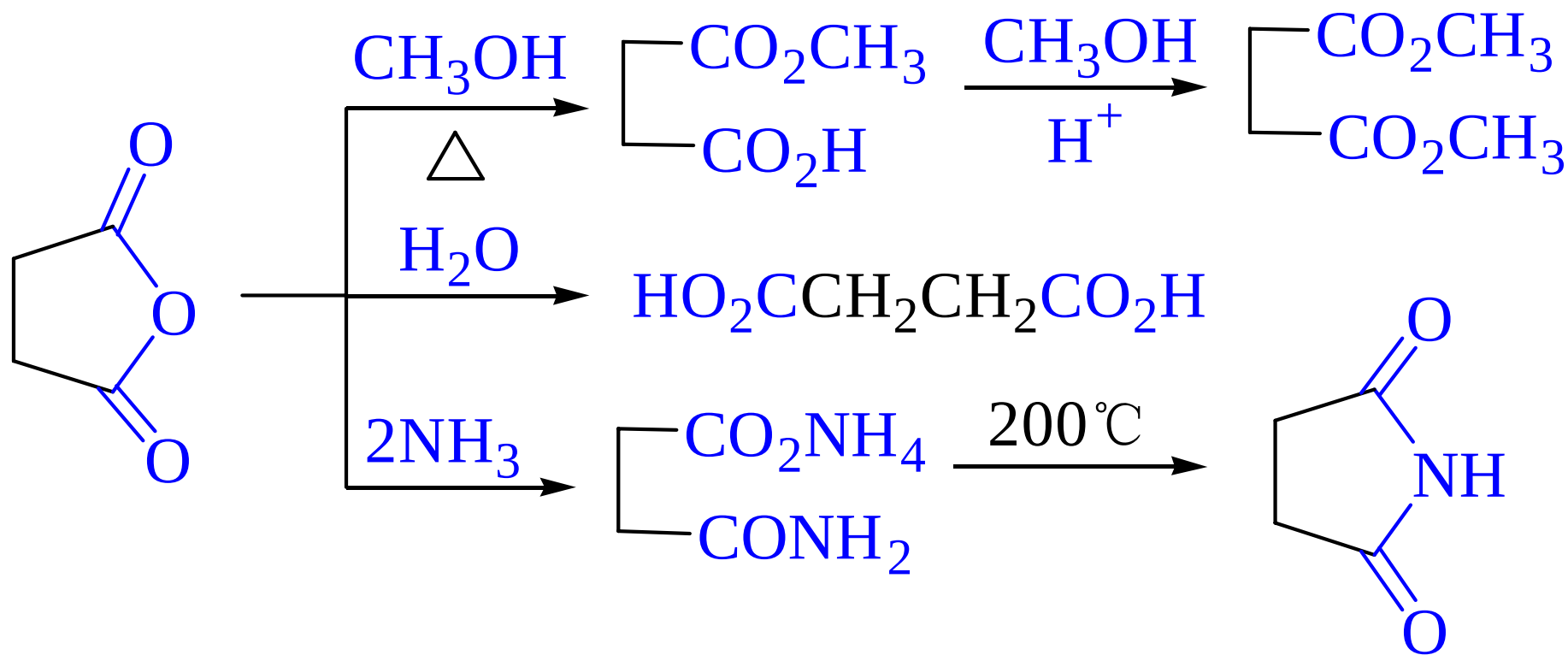
12.5.1 亲核取代反应

3. 氨解反应

- 羧酸衍生物氨解都得到酰胺
- 反应在碱性条件下进行较为有利
- 除3°胺外的胺均能与羧酸衍生物发生氨解反应

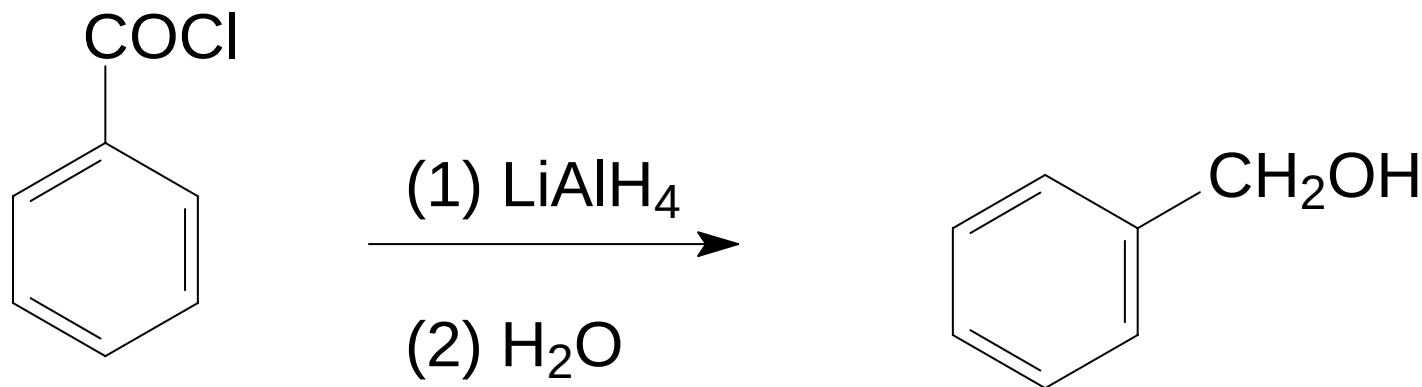
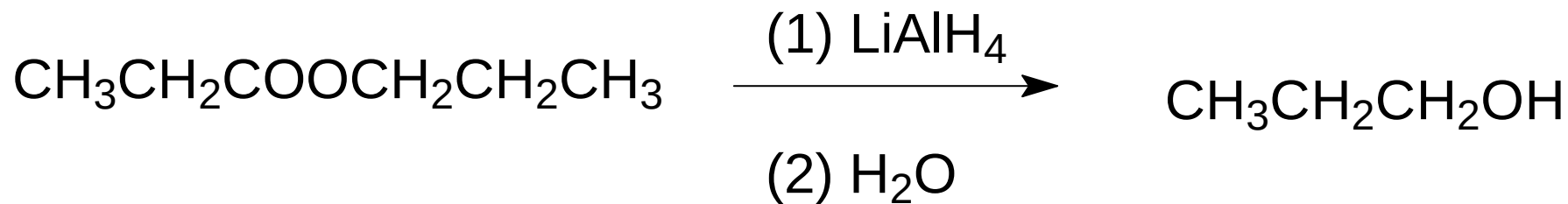


丁二酸酐的三解实例

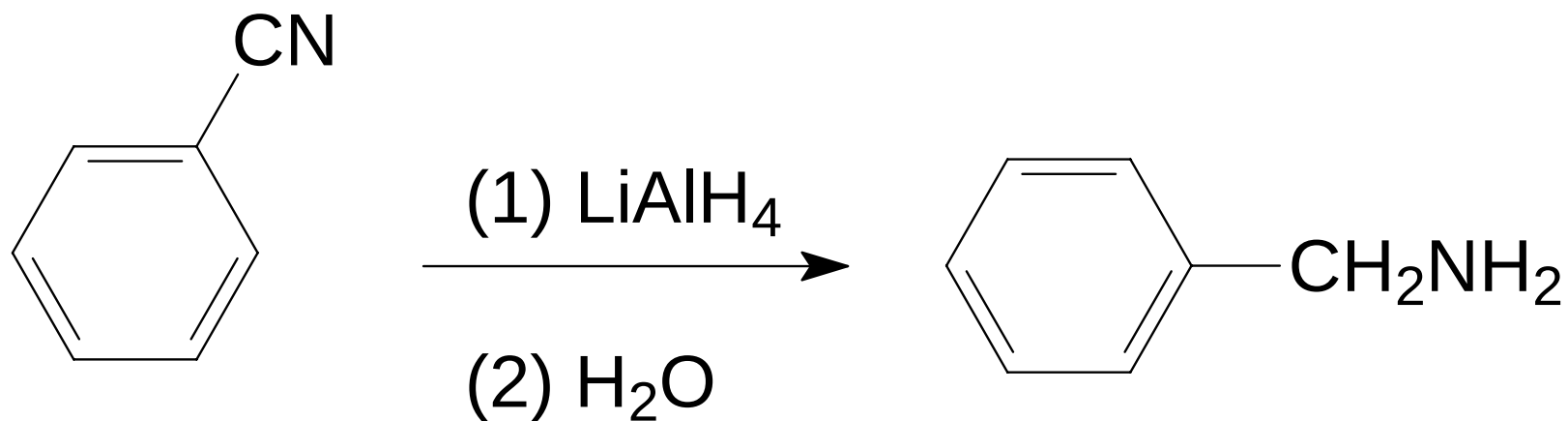
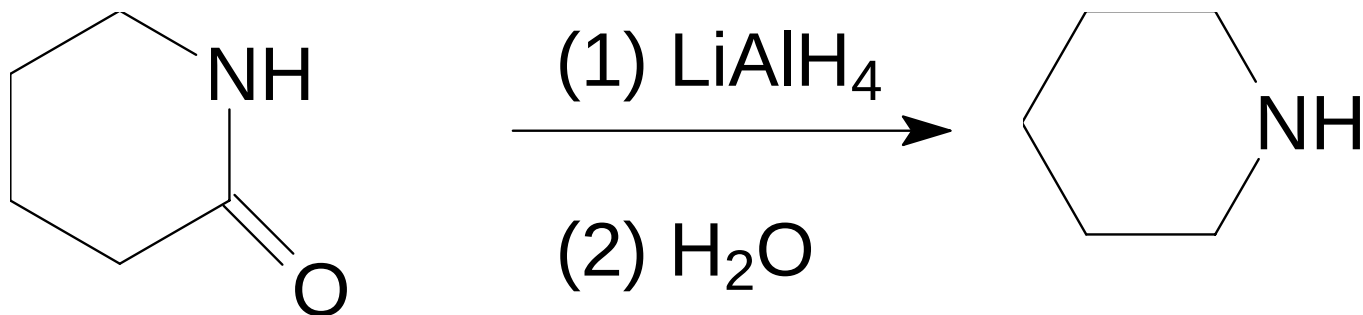


12.5.2 羧酸衍生物的还原反应

1. 用金属氢化物还原 (LiAlH_4)

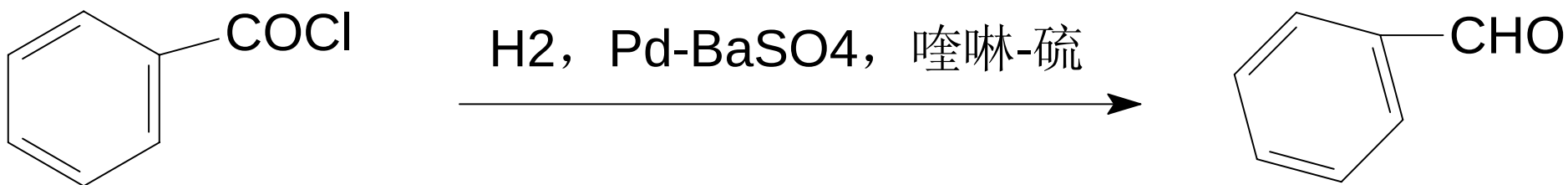


Example

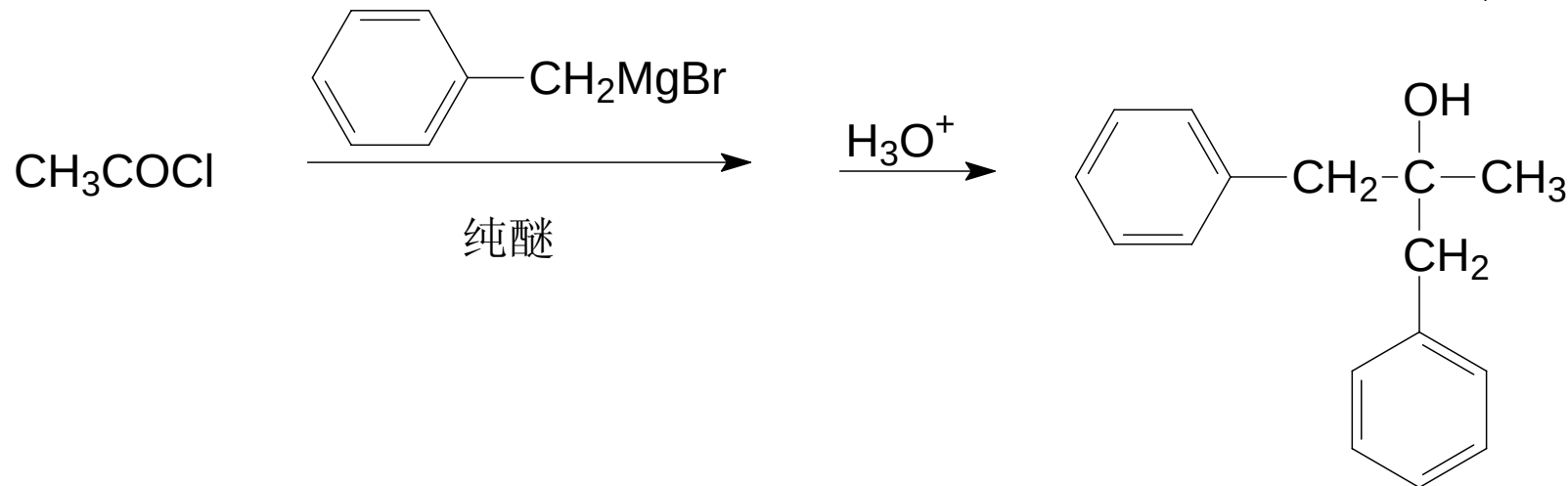
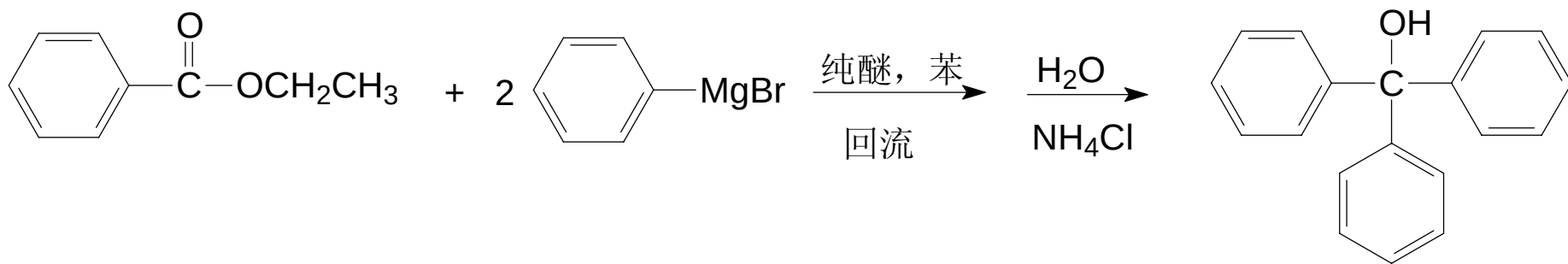
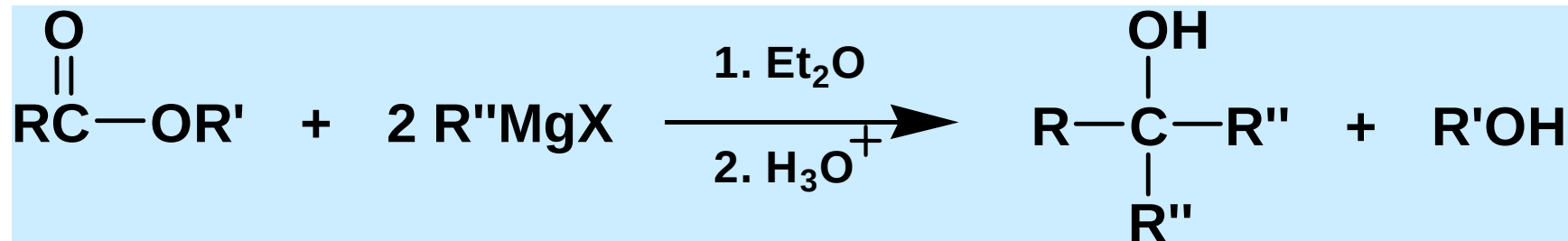


12.5.2 羧酸衍生物的还原反应

2. Rosenmund还原

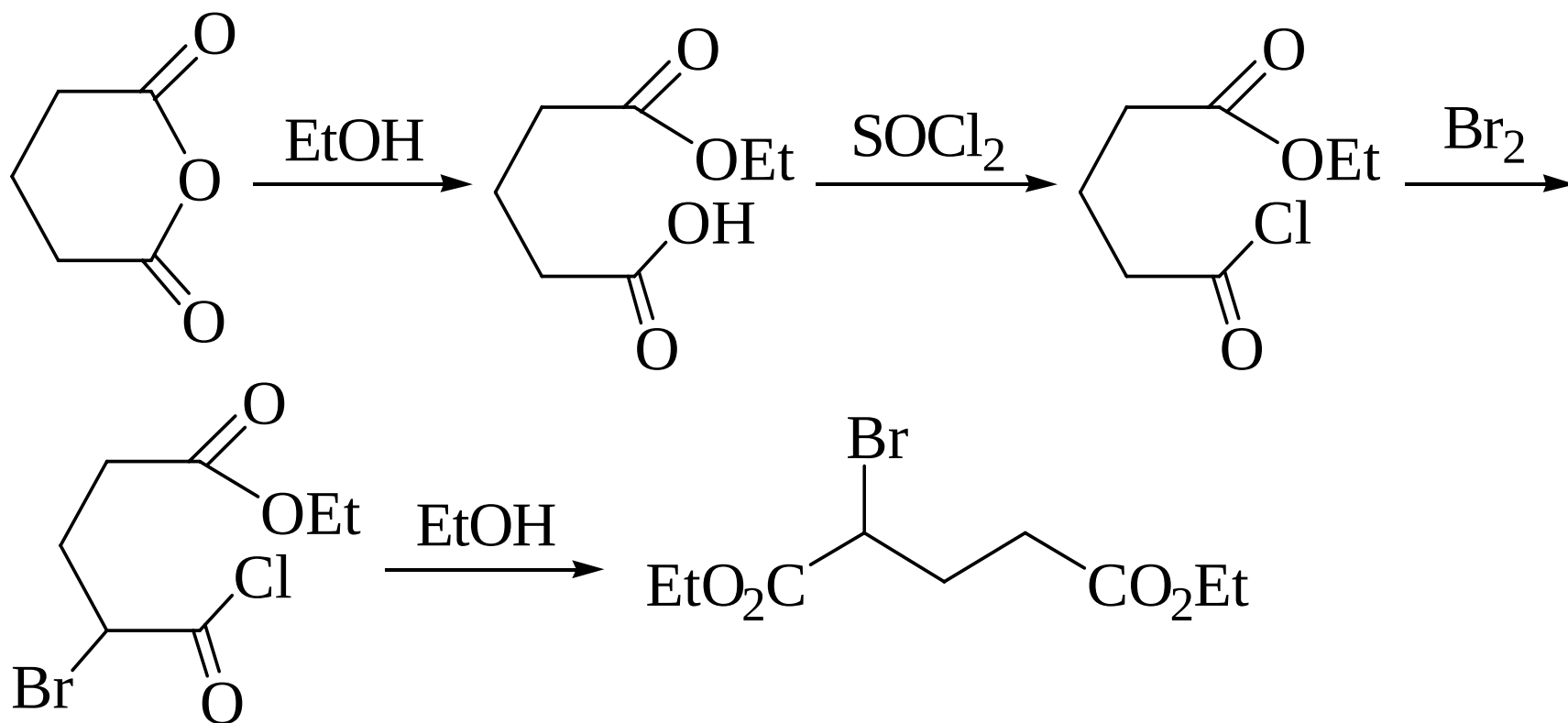


12.5.3 与有机金属化合物的反应



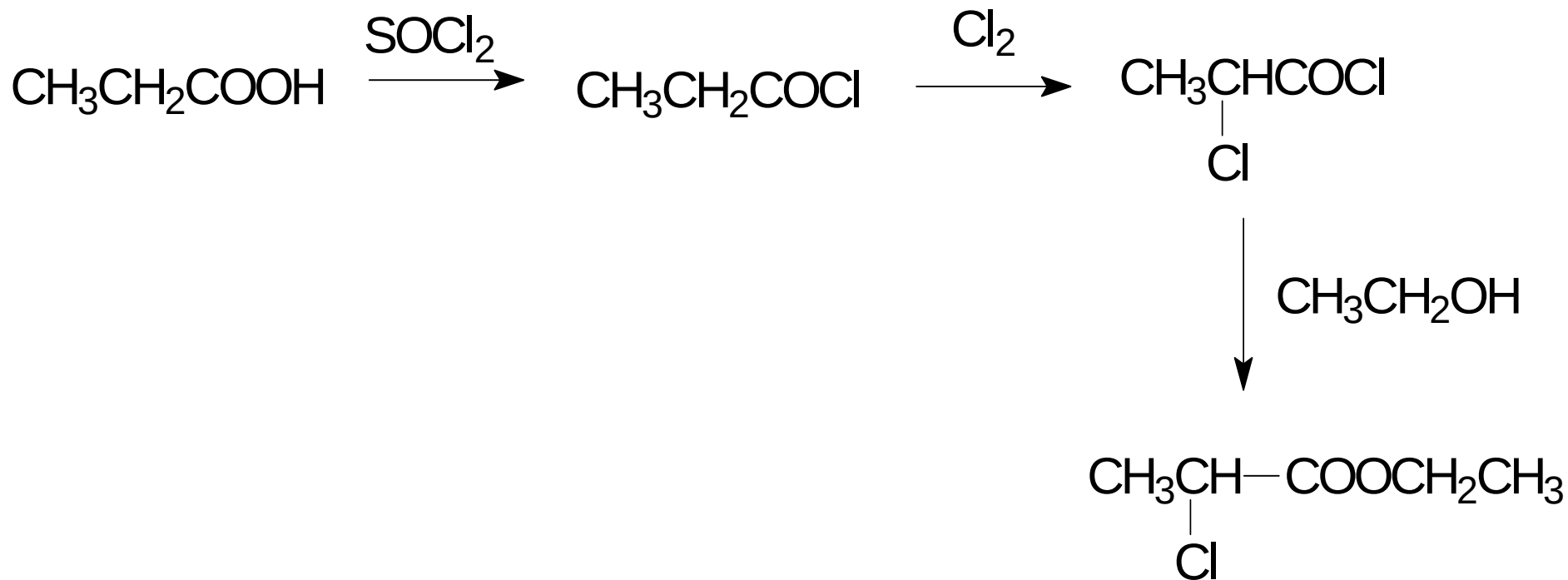
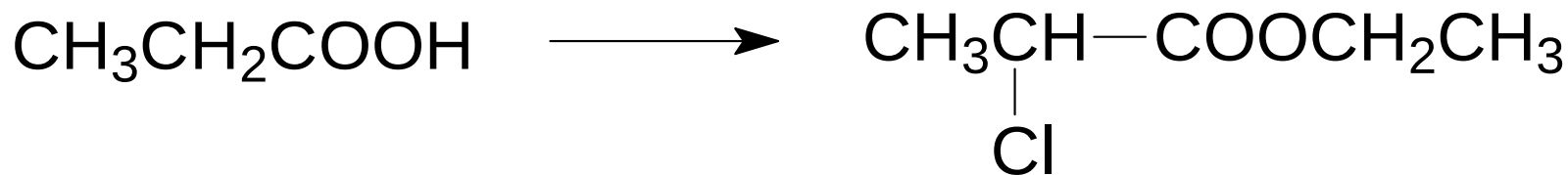
厦门大学有机化学B

12.5.4 酰卤的 α -卤代



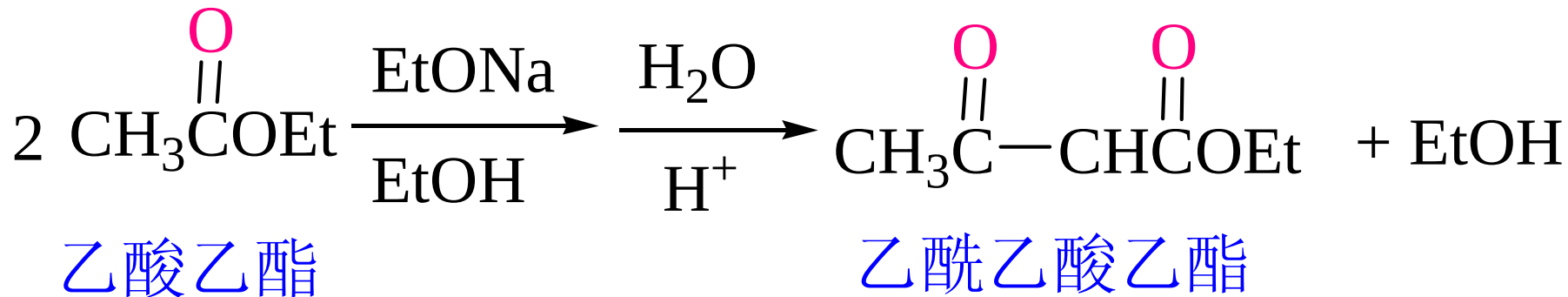
Example

请同学们完成下列转化

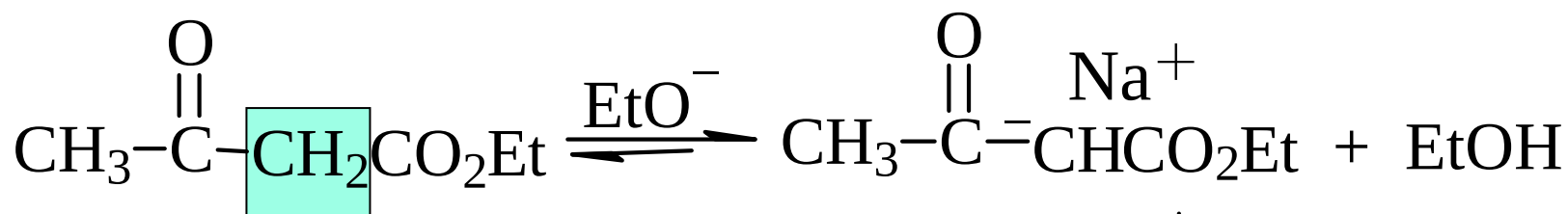
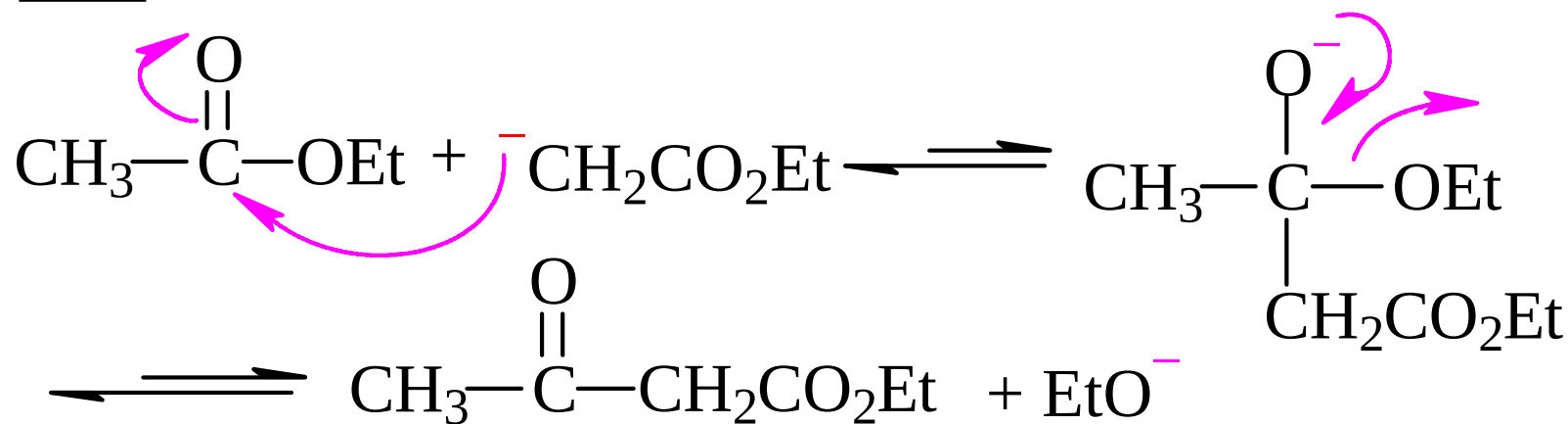
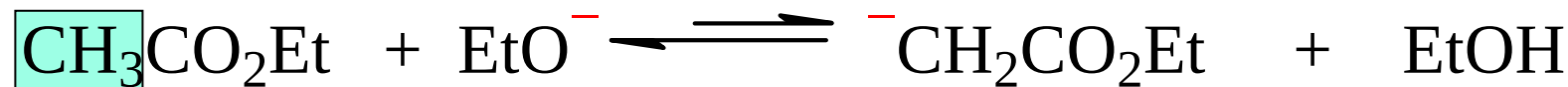


12.5.5 酯缩合反应

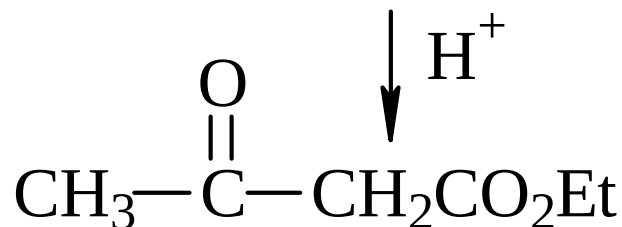
Claisen 酯缩合：含 α -氢的酯在碱的作用下，两分子酯发生缩合反应，生成 β -羰基酸酯同时消去一分子醇



pKa 24.5

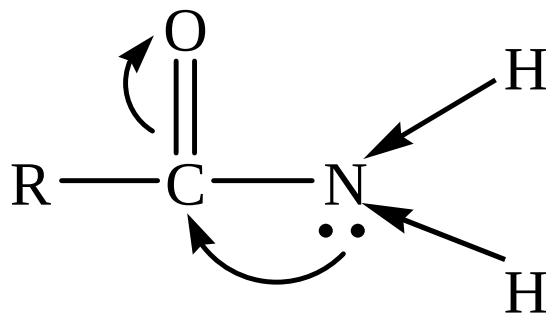
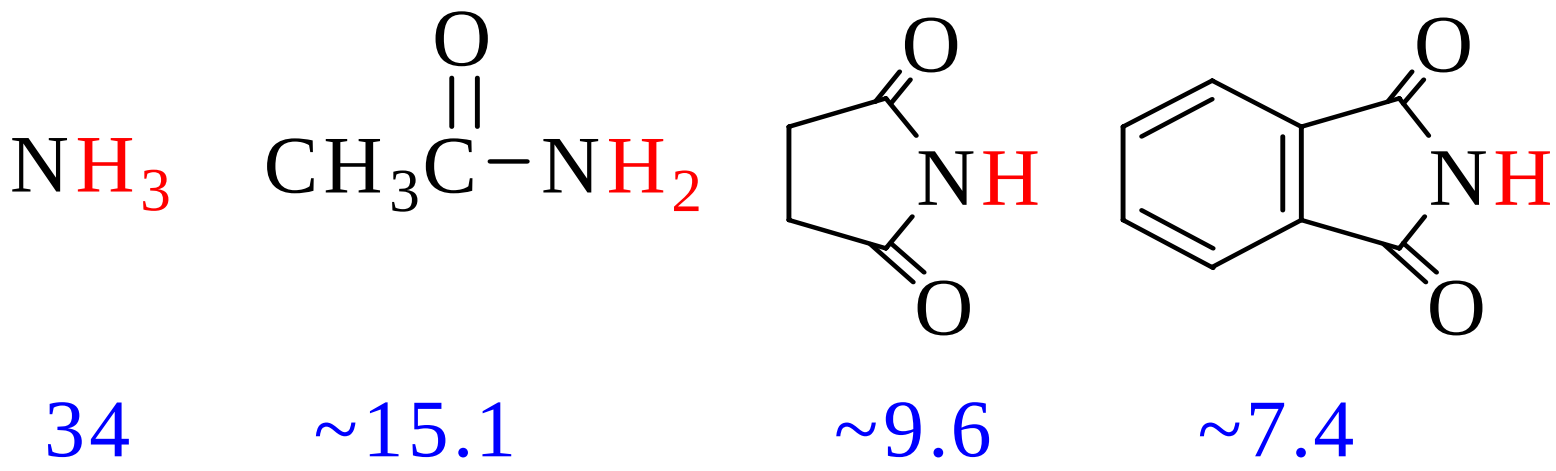


pKa ~11

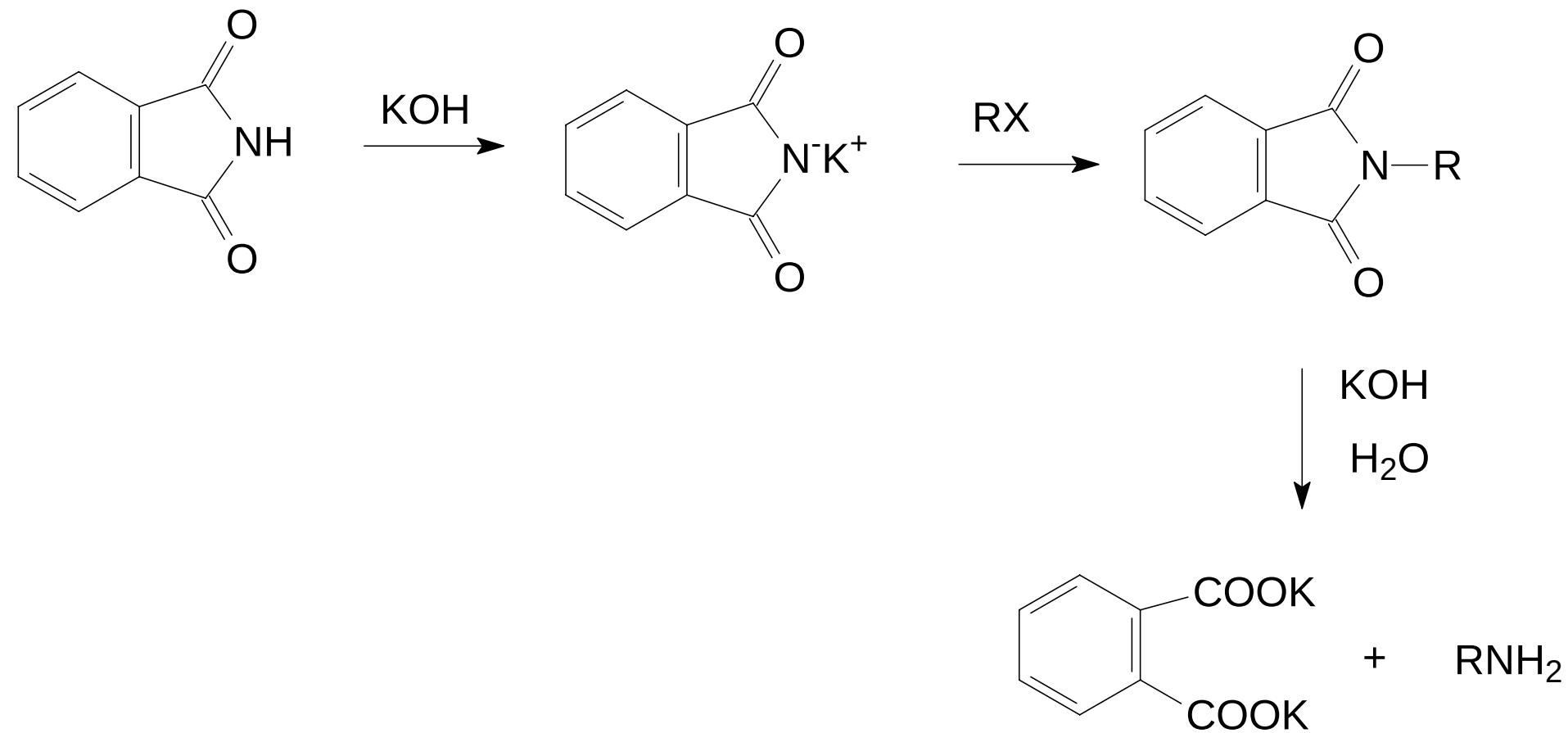


12.6 酰胺的特殊性质

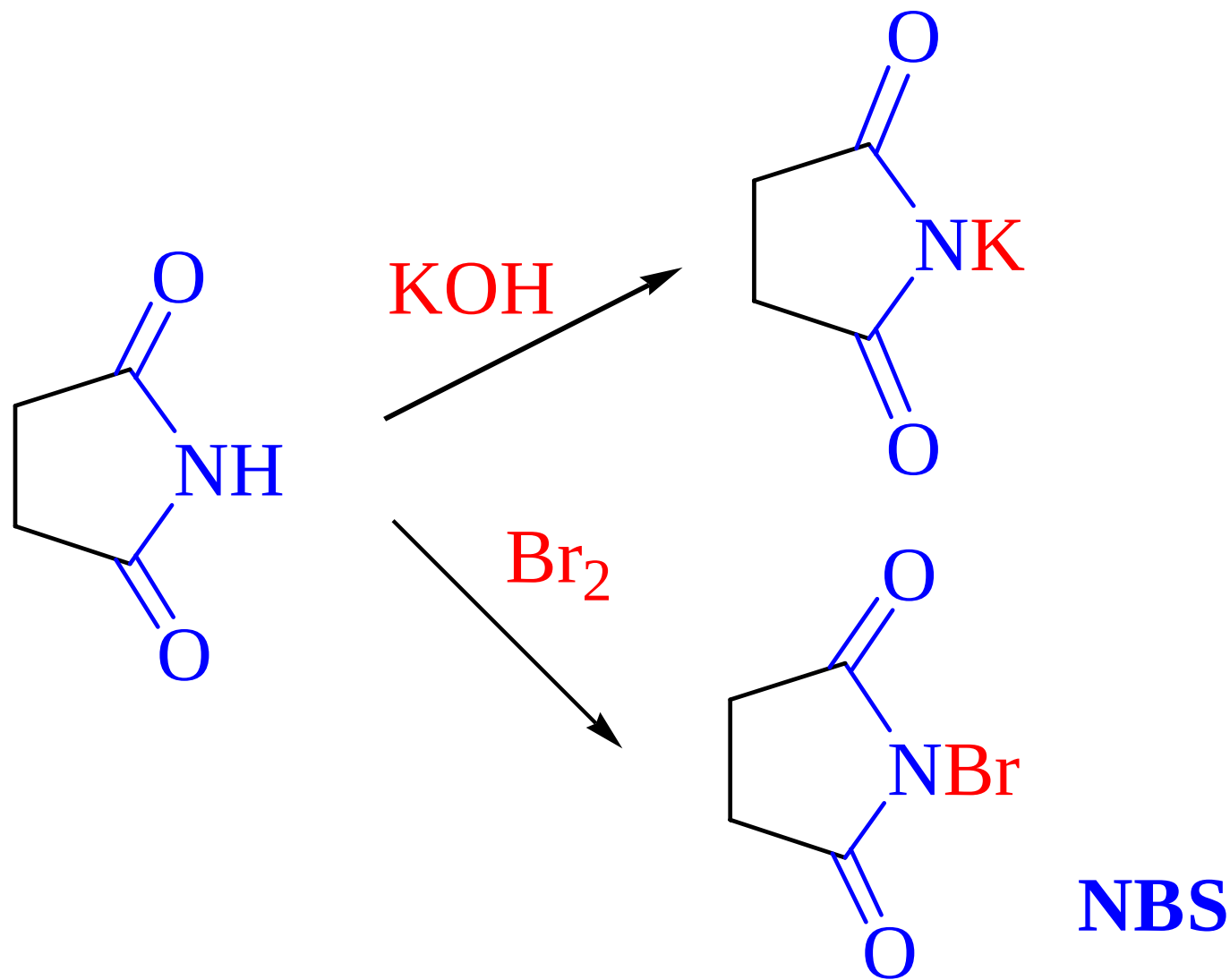
1. 酰胺的酸碱性



Gabriel反应----用于伯胺的制备



丁二酰亚胺的反应

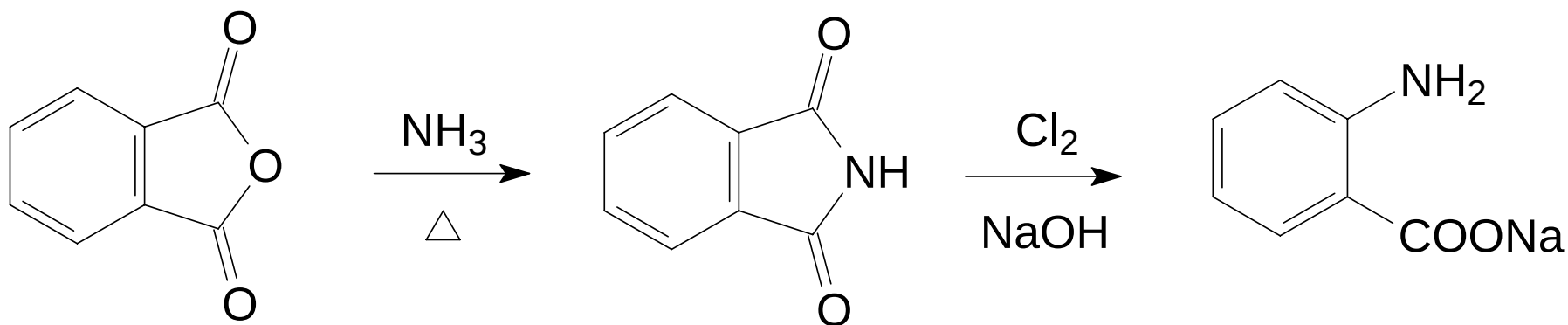
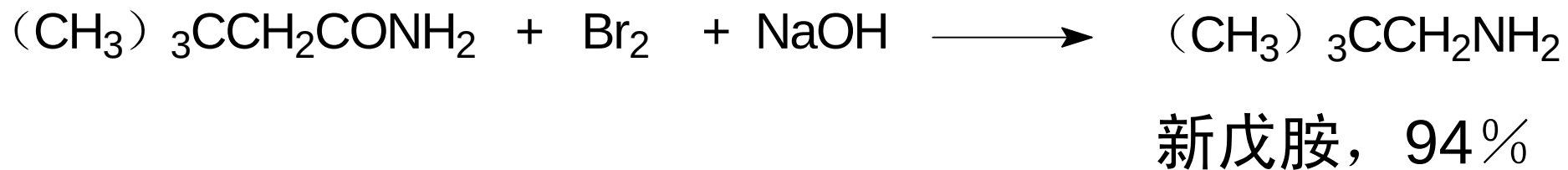


12.6 酰胺的特殊性质

2. Hofmann降解反应

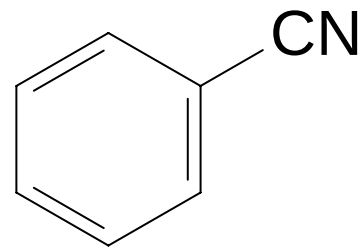
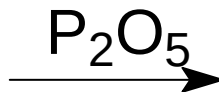
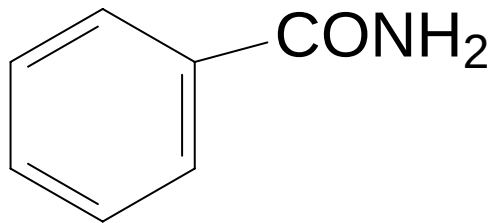
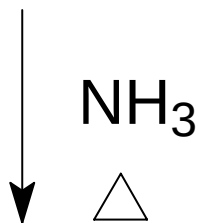
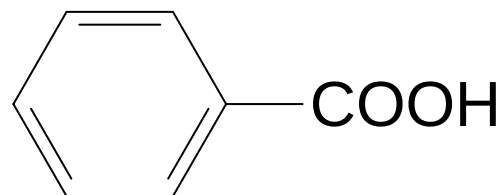
酰胺与溴或氯在碱溶液中作用，脱去羰基生成伯胺，使碳链减少一个碳原子的反应，通常称为Hofmann降解反应





12.6 酰胺的特殊性质

3. 酰胺的脱水



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作业

- P256: 12.2
- P257: 12.3
- P258: 12.5
- P259: 12.6

习题

P279: 3,
4(1)(2)(3)(4)(5),
6, 10

- P267: 12.10
- P268: 12.11
- P270: 12.12

附录

知识储备（自学）

Some Carboxylic Acids

Structure	Common name	
HCOOH	Formic acid	蚁酸
CH_3COOH	Acetic acid	醋酸
$\text{CH}_3\text{CH}_2\text{CH}_2\text{COOH}$	Butyric acid	酪酸
$\text{CH}_3(\text{CH}_2)_{10}\text{COOH}$	Lauric acid	月桂酸
$\text{CH}_3(\text{CH}_2)_{14}\text{COOH}$	Palmitic acid	棕榈酸, 软脂酸
$\text{CH}_3(\text{CH}_2)_{16}\text{COOH}$	Stearic acid	硬脂酸
$\text{CH}_2=\text{CHCOOH}$	Acrylic acid	丙烯酸
$\text{C}_6\text{H}_5\text{COOH}$	Benzoic acid	苯甲酸

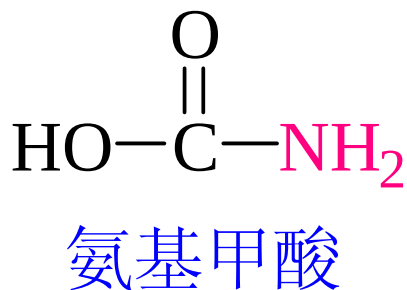
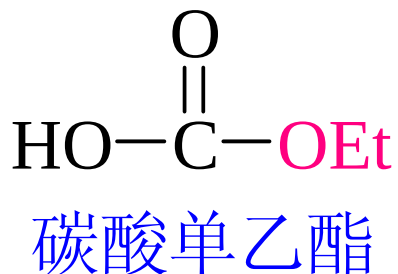
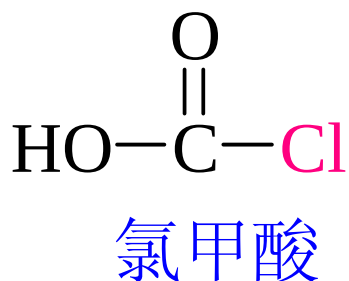
Some Carboxylic Acids

Structure	Common name	
HOOCCOOH	Oxalic acid	草酸
$\text{HOOC}\text{CH}_2\text{COOH}$	Malonic acid	缩苹果酸
$\text{HOOC}(\text{CH}_2)_2\text{COOH}$	Succinic acid	琥珀酸
$\text{HOOC}(\text{CH}_2)_4\text{COOH}$	Adipic acid	肥酸
$\text{HOOC}\text{CH}=\text{CH}\text{COOH}(\text{cis})$	Maleic acid	马来酸
$\text{HOOC}\text{CH}=\text{CH}\text{COOH}(\text{trans})$	Fumaric acid	富马酸
$\text{o-C}_6\text{H}_4(\text{COOH})_2$	Phthalic acid	酞酸

碳酸、原酸、黄原酸及其衍生物

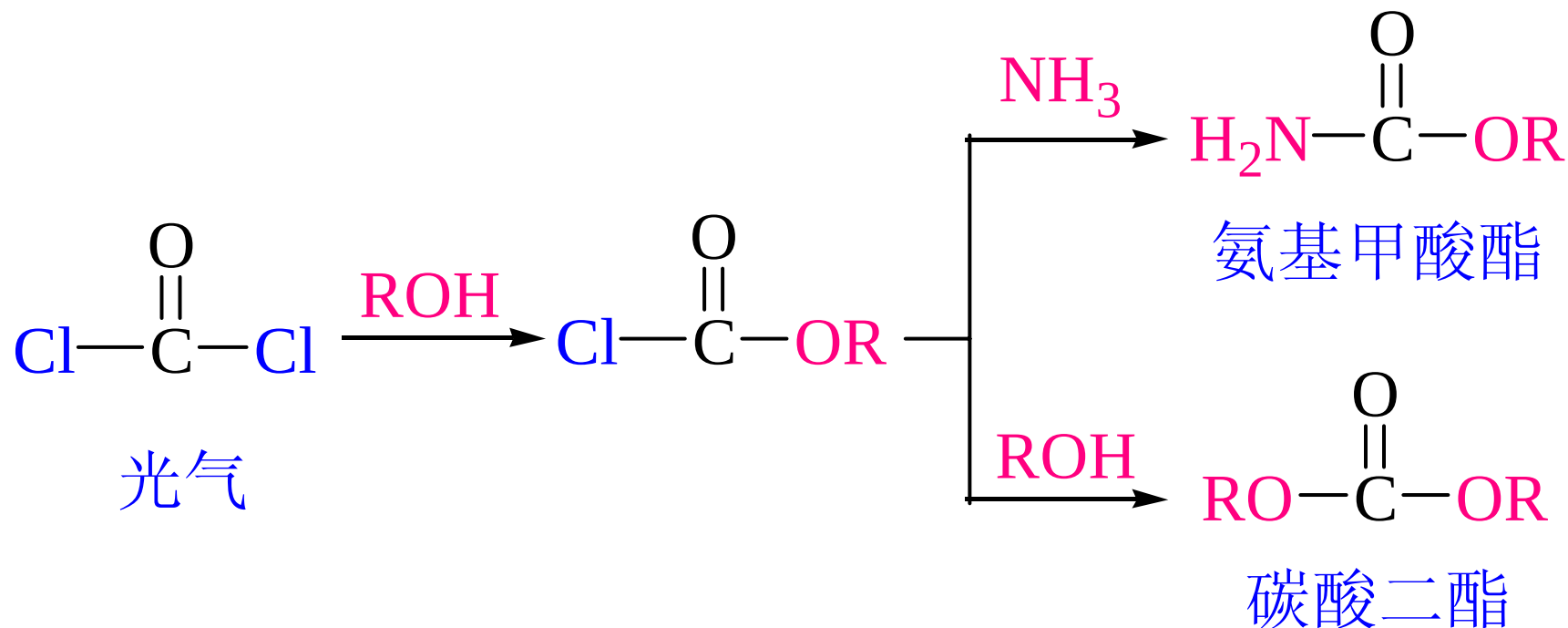


碳酸的单酰氯、单酯、单酰胺都不大稳定，易失去二氧化碳

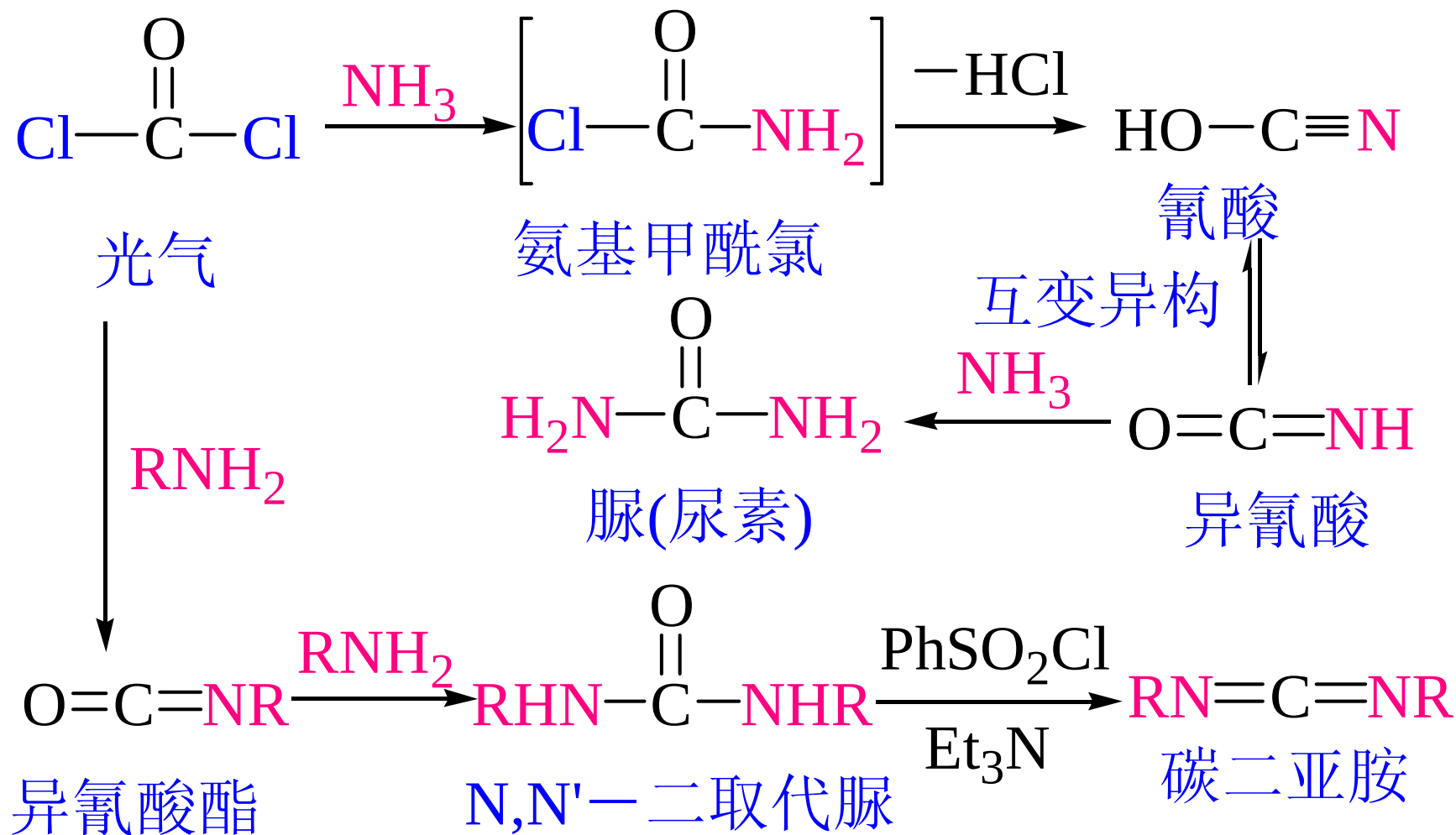


碳酸的衍生物

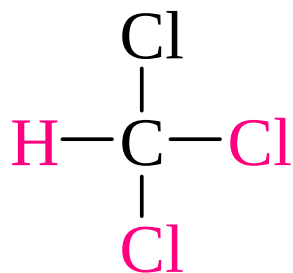
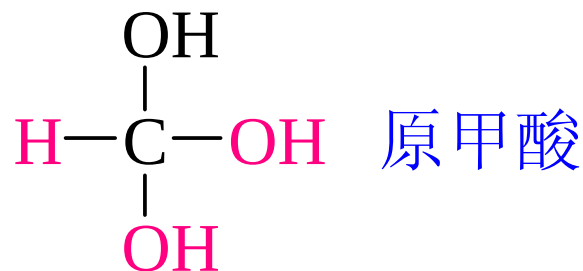
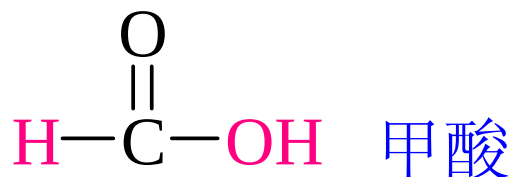
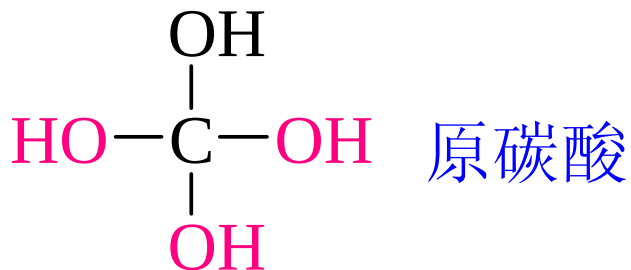
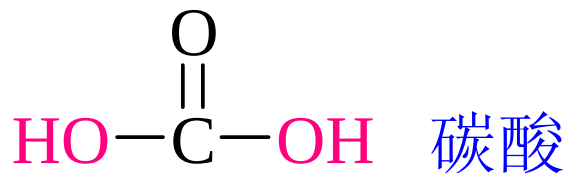
碳酸二个羟基被取代所形成的衍生物是稳定的



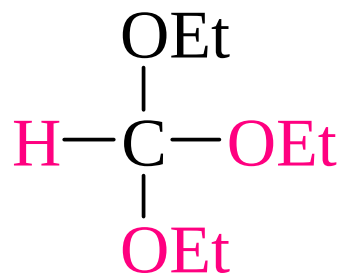
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原酸及其衍生物



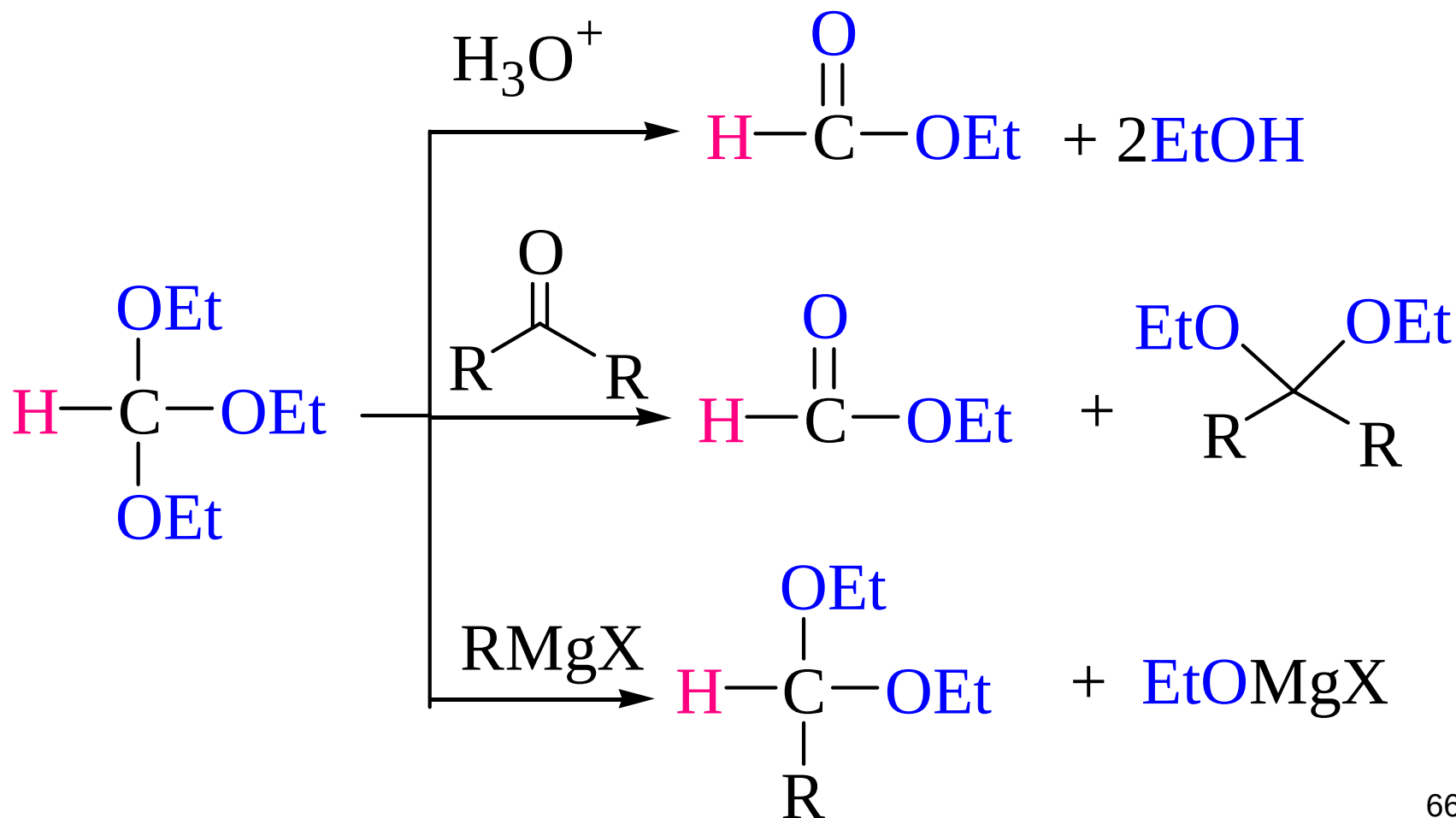
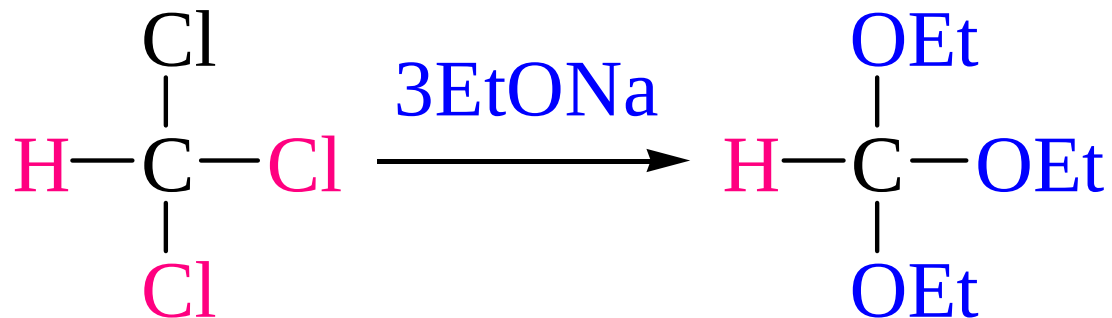
原甲酸三酰氯(氯仿)



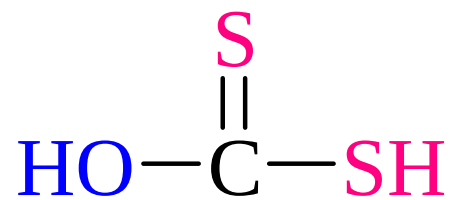
原甲酸三乙酯

原酸可看成是羧酸的羰基水合的产物。

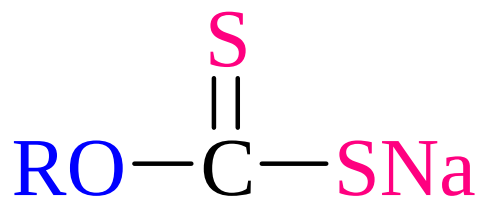
原酸不稳定，但原酸的三酰氯、三酯是稳定的。



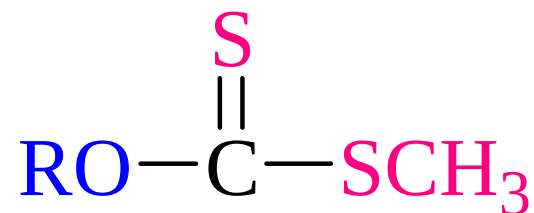
黄原酸及其衍生物



黄原酸



烷基黄原酸钠



烷基黄原酸甲酯

